2.32 Cut-back method of attenuation measurement Cut-back method is a destructive measurement technique for determining the attenuation α of a fiber. The first part of the experiment involves measuring the optical power P_{far} coming out from the fiber at the far end as shown in Figure 2.54. Then, in the second part, keeping everything the same, the fiber is cut close to the launch or the source end. The output power P_{near} is measured at the near end from the shortcut fiber. The attenuation is then given by

$$\alpha = (-10/L) \log (P_{\text{far}}/P_{\text{near}})$$

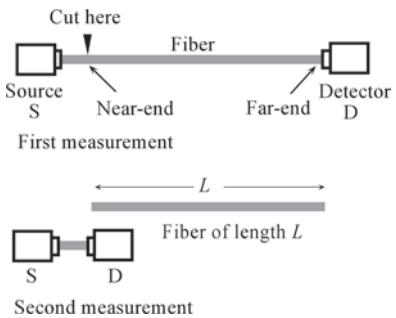


FIGURE 2.54 Illustration of the cut-back method for measuring the fiber attenuation. S is an optical source and D is a photodetector.

in which L is the separation of the measurement points, the length of the cut fiber, and α is in dB per unit length. The output P_{near} from the shortcut fiber in the second measurement is actually the input into the fiber under test in the first experiment. Usually a mode scrambler (mode stripper) is used for multimode fibers before the input. The power output from a particular fiber is measured to be 13 nW. Then, 10 km of fiber is cut out and the power output is measured again and found to be 43 nW. What is the attenuation of the fiber?

2.33 Intrinsic losses

(a) Consider a silica-based standard single-mode fiber with a NA of 0.15. What is its attenuation at 1550 nm? How does this compare with the attenuation quotes for the Corning SMF@28e$^+$, 0.16–0.19 dB km^{-1}at 1550 nm?

(b) Consider a step index fiber $n_1 = 1.47$ and $n_2 = 1.46$, core diameter $= 50$ μm. Corresponding to the maximum value of the incident angle for which light rays will be guided through this fiber, calculate the number of reflections and the loss that would take place in one kilometre length of the fiber assuming 0.01% loss of power at each reflection.

2.34 Scattering losses and fictive temperature Rayleigh scattering process decreases with wavelength, and as mentioned in Chapter 1, it is inversely proportional to λ^4. The expression for the attenuation α_R in a single component glass such as silica due to Rayleigh scattering is approximately given by two sets of different equations in the literature[32]

$$\alpha_R \approx \frac{8\pi^3}{3\lambda^4} n^8 p^2 \beta_T k_B T_f \quad \text{and} \quad \alpha_R \approx \frac{8\pi^3}{3\lambda^4} (n^2 - 1)^2 \beta_T k_B T_f$$

in which λ is the free-space wavelength, n is the refractive index at the wavelength of interest, β_T is the isothermal compressibility (at T_f) of the glass, k_B is the Boltzmann constant, and T_f is a quantity called the *fictive temperature* (or the glass transition temperature) at which the liquid structure during the cooling of the fiber is frozen to become the glass structure. Fiber is drawn at high temperatures. As the fiber cools, eventually the temperature drops sufficiently for the atomic motions to be so sluggish that the structure becomes essentially "frozen-in" and remains like this even at room temperature. Thus, T_f marks the temperature below which the liquid structure is frozen and hence the density fluctuations are also frozen into the glass structure. Use these two equations and calculate the attenuation in dB km^{-1} due to Rayleigh scattering at $\lambda = 1.55\,\mu$m given that pure silica (SiO_2) has the following properties: $T_f \approx 1180°C$; $\beta_T \approx 7 \times 10^{-11}\,\text{m}^2\,\text{N}^{-1}$ (at high temperatures); $n = 1.5$ at 1.5 μm; $p = 0.28$. The lowest reported attenuation around this wavelength is about 0.14 dB km^{-1}. What is your conclusion? (Note that α_B above is in Nepers per meter.)

2.35 Bending loss Bending losses always increase with the mode field diameter. Since the MFD increases for decreasing V, $2w \approx 2 \times 2.6a/V$, smaller V fibers have higher bending losses. How does the bending loss

[32] For example, R. Olshansky, *Rev. Mod. Phys.*, *51*, 341, 1979.

α vs. radius of curvature R behavior look like on a semilogarithmic plot [as in Figure 2.39 (a)] for two values of the V-number V_1 and V_2 if $V_2 > V_1$? It is found that for a single-mode fiber with a cutoff wavelength $\lambda_c = 1180\,\text{nm}$, operating at 1300 nm, the microbending loss reaches 1 dB m^{-1} when the radius of curvature of the bend is roughly 6 mm for $\Delta = 0.00825$, 12 mm for $\Delta = 0.00550$, and 35 mm for $\Delta = 0.00275$. Explain these findings.

2.36 Bend loss reduced fibers Consider the bend loss measurements listed in Table 2.8 for four different types of fiber. The trench fibers have a trench placed in the cladding where the refractive index is lowered as shown in Figure 2.39. The nanoengineered fiber is shown in Figure 2.55. There is a ring of region in the cladding in which there are gas-filled nanoscale voids. (They are introduced during fabrication.) A void in the ring has a circular cross-section but has a length along the fiber that can be a few meters. These voids occupy a volume in the ring that is only 1–10%. Plot the bending loss semilogarithmically (α on a log scale and R on a linear scale) and fit the data to $\alpha_{\text{micobend}} = A\exp(-R/R_c)$ and find A and R_c. What is your conclusion? Suppose that we set our maximum acceptable bending loss to 0.1 dB turn^{-1} in installation (the present goal is to bring the bending loss to below 0.1 dB turn^{-1}). What are the allowed radii of curvature for each turn?

TABLE 2.8 Bend radius R in mm, α in dB turn^{-1}. Data over 1.55–1.65 μm range

Standard SMF[a] 1550 nm		Trench Fiber 1[b] 1650 nm		Trench Fiber 2[c] 1625 nm		Nanoengineered Fiber[a] 1550 nm	
R (mm)	α (dB turn^{-1})	R (mm)	α (dB turn^{-1})	R (mm)	α (dB turn^{-1})	R (mm)	α (dB turn^{-1})
5.0	15.0	7.50	0.354	5.0	0.178	5.0	0.031
7.0	4.00	10.0	0.135	7.5	0.0619	7.5	0.0081
10.0	0.611	15.0	0.020	10.0	0.0162	10.0	0.0030
12.5	0.124			15.0	0.00092	15.0	0.00018
16.0	0.0105						
17.5	0.0040						

(*Source:* Data used from a number of sources: [a]M. -J. Li *et al.*, *J. Light Wave Technol.*, *27*, 376, 2009; [b]K. Himeno *et al.*, *J. Light Wave Technol.*, *23*, 3494, 2005; [c]L. -A. de Montmorillon, *et al.* "Bend-Optimized G.652D Compatible Trench-Assisted Single-Mode Fibers," *Proceedings of the 55th IWCS/Focus*, pp. 342–347, November, 2006.)

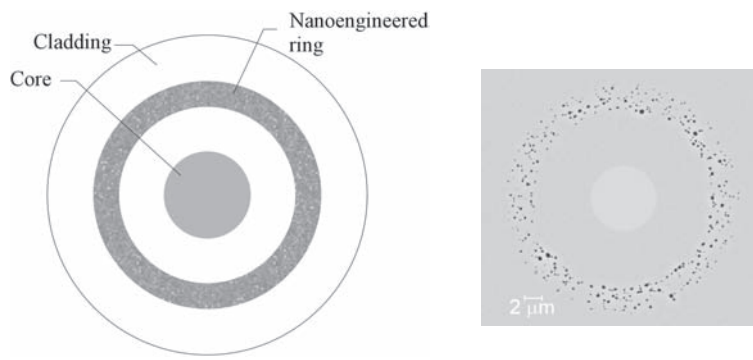


FIGURE 2.55 Left: The basic structure of bend-insensitive fiber with a nanoengineered ring in the cladding. Right: An SEM picture of the cross-section of a nanoengineered fiber with reduced bending losses. *(Courtesy of Ming-Jun Li, Corning Inc. For more information see U.S. Patent 8,055.110, 2011.)*

2.37 Microbending loss Microbending loss α_B depends on the fiber characteristics and wavelength. We will calculate α_B approximately given various fiber parameters using the single-mode fiber microbending loss equation (D. Marcuse, *J. Op. Soc. Am.*, *66*, 216, 1976)

$$\alpha_B \approx \frac{\pi^{1/2}\kappa^2}{2\gamma^{3/2}V^2\left[K_1(\gamma a)\right]^2} R^{-1/2} \exp\left(-\frac{2\gamma^3}{3\beta^2} R\right)$$

where R is the bend radius of curvature; a is the fiber radius; β is the propagation constant, determined by b, normalized propagation constant, which is related to V, $\beta = n_2k\left[1 + b\Delta\right]$; $k = 2\pi/\lambda$ is the free-space wave vector; $\gamma = [\beta^2 - n_2^2k^2]^{1/2}$; $\kappa = [n_1^2k^2 - \beta^2]^{1/2}$; and $K_1(x)$ is a first-order modified Bessel function, available in math software packages. The normalized propagation constant b can be found from $b = (1.1428 - 0.996V^{-1})^2$. Consider a single-mode fiber with $n_1 = 1.450, n_2 = 1.446, 2a$ (diameter) $= 3.9\,\mu$m. Plot α_B vs. R for $\lambda = 633$ nm and 790 nm from $R = 2$ mm to 15 mm. Figure 2.56 shows the experimental results on an SMF that has the same properties as the fiber above. While the data in Figure 2.56 are for a 10 cm fiber, the equation above gives the attenuation per meter. We need to divide the calculated values by 10 to compare the two sets of data. What is your conclusion? (Note that while the data in Figure 2.56 are for a 10 cm fiber, the equation above gives the attenuation per meter. We need to divide the calculated attenuation values by 10 to compare the two sets of data. You might wish to compare your calculations with the experiments of A. J. Harris and P. F. Castle, *IEEE J. Light Wave Technol., LT4*, 34, 1986).

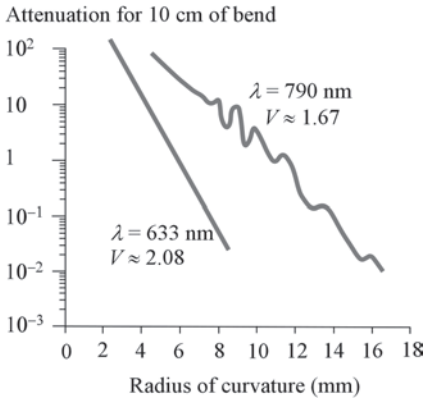


FIGURE 2.56 Measured microbending loss (attenuation) for a 10-cm fiber bent by different amounts of radius of curvature R. Single-mode fiber with a core diameter of 3.9 μm, cladding radius of 48 μm, $\Delta = 0.00275$, NA ≈ 0.10, $V \approx 1.67$ and 2.08. (*Source:* Data extracted from A. J. Harris and P. F. Castle, "Bend Loss Measurements on High Aperture Single-Mode Fibers as a Function of Wavelength and Bend Radius," *IEEE J. Light Wave Technol.*, *LT14*, 34, 1986, and replotted with a smoothed curve; [see original article for the discussion of peaks in α_B vs. R at 790 nm]).

2.38 Fiber Bragg Grating (FBG) A silica fiber-based FBG is required to operate at 850 nm. What should be the periodicity of the grating Λ? If the amplitude of the index variation Δn is 2×10^{-5} and total length of the FBG is 5 mm, what are the maximum reflectance at the Bragg wavelength and the bandwidth of the FBG? Assume that the effective refractive index $\overline{n}$ is 1.460. What are the reflectance and the bandwidth if Δn is 2×10^{-4}?



Fiber Bragg Grating (FBG)–based optical strain sensor for monitoring strains in civil structures, for example, bridges, dams, buildings, and tunnels. The fiber is mounted on a steel carrier and maintained stretched. *(Courtesy of Micron Optics.)*

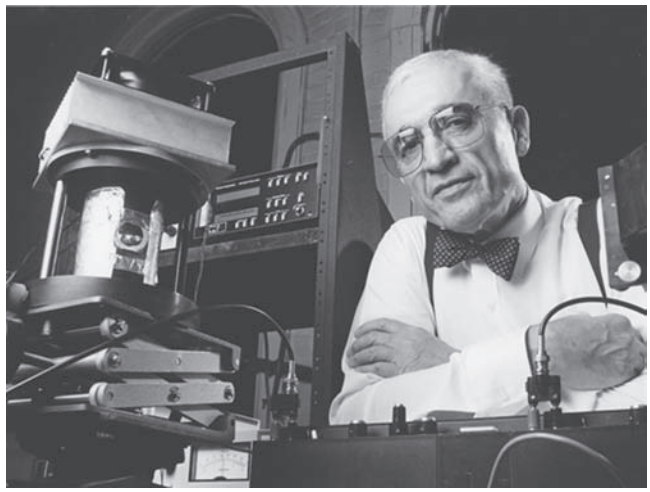
2.39 Fiber Bragg Grating sensor array Consider a FBG strain sensor array embedded in a silica fiber that is used to measure strain at various locations on an object. Two neighboring sensors have grating periodicities of $\Lambda_1 = 534.5$ nm and $\Lambda_2 = 539.7$ nm. The effective refractive index is 1.450 and the photoelastic coefficient is 0.22. What is the maximum strain that can be measured? What would be the main problem with this sensor array? What is the strain at fracture if the fiber fractures roughly at an applied stress of 700 MPa and the elastic modulus is 70 GPa? What is your conclusion?



Fiber Bragg Grating (FBG)–based optical temperature sensors for use from −200°C to 250°C. FBGs are mounted in different packages: top left, all dielectric; top right, stainless steel; and bottom, copper. The sensors operate over 1510–1590 nm. *(Courtesy of Lake Shore Cryotronics Inc.)*

I wanted visible light, because I knew that if I could get red light, other colors would be possible.

—Nick Holonyak Jr.



Nick Holonyak Jr. carried out the early work in the development of practical light-emitting diodes (LEDs) in the visible spectrum during the 1960s while working as a consulting research scientist for General Electric Co. in Syracuse. He made his first visible laser-LED in 1962, which emitted red light.[1] It served as the basis for the concept of the "ultimate lamp." Since 1963, he has been at the University of Illinois at Urbana-Champaign, where he currently holds the John Bardeen Endowed Chair. In the February 1963 issue of *Readers Digest*, Nick Holonyak Jr. suggested that the incandescent light bulb will eventually be replaced by the LED. The above photo was taken around 1970–1975. *(Courtesy of University of Illinois at Urbana-Champaign.)*

[1] N. Holonyak Jr. and S. F. Bevacqua, "Coherent (Visible) Light Emission from a Ga(As_xP_{1-x}) Junction," *Appl. Phys. Letts., 1*, 82, 1962.

3 SEMICONDUCTOR SCIENCE AND LIGHT-EMITTING DIODES

3.1 REVIEW OF SEMICONDUCTOR CONCEPTS AND ENERGY BANDS

A. Energy Band Diagrams, Density of States, Fermi–Dirac Function and Metals

We know from modern physics that the energy of the electron in an atom is quantized and can only have certain discrete values as illustrated for the Li atom in Figure 3.1. The isolated Li atom has two electrons in the 1*s* shell and one electron in the 2*s* subshell. The same concept also applies to the electron energy in a molecule with several atoms. Again the electron energy is quantized. However, when we bring together something like 10^{23} Li atoms to form the metal crystal, the interatomic interactions result in the formation of electron **energy bands**. The 2*s* energy level splits into some 10^{23} closely spaced energy levels that effectively form an **energy band**, which is called the 2*s* band. Similarly other higher energy levels also form bands as illustrated in Figure 3.1. These energy bands overlap to form one continuous energy band which represents the energy band structure of a metal. The 2*s* energy level in the Li atom is half full (2*s* subshell needs 2 electrons but has 1) which means that the 2*s* band in the crystal will also be half full. Metals characteristically have partially filled energy bands.

In the case of metals, the energy band diagram is straightforward and essentially one band of energies in which the energy levels from the bottom of the band to a certain level are all filled at zero Kelvin (0 K), as indicated in Figure 3.1. Each energy level in this band corresponds to a wavefunction for the electron inside the crystal, and represents a **quantum state** for the electron. Thus, the energy levels are often referred to as energy states. In the solid, the bottom of the band is normally taken as the zero of energy so that electrons fill all the states until a certain energy called the **Fermi energy**, denoted as E_F—the highest occupied energy level at 0 K. The states above E_F are empty up to the vacuum level. The energy required to remove an electron from the metal is the energy involved in taking an electron from E_F to the vacuum level, which is called the **work function** Φ of the metal. For example, a photon with an energy $h\upsilon$ greater than Φ that is incident on a metal can cause an electron at E_F to become ejected out from the metal to become free, a phenomenon called the **photoelectric effect**. The electron is said to be **photoemitted.** Many photomultiplier tubes in optoelectronics have a cathode material with the right Φ that allows electrons to be photoemitted when light within the desired range of wavelengths is incident on the cathode.

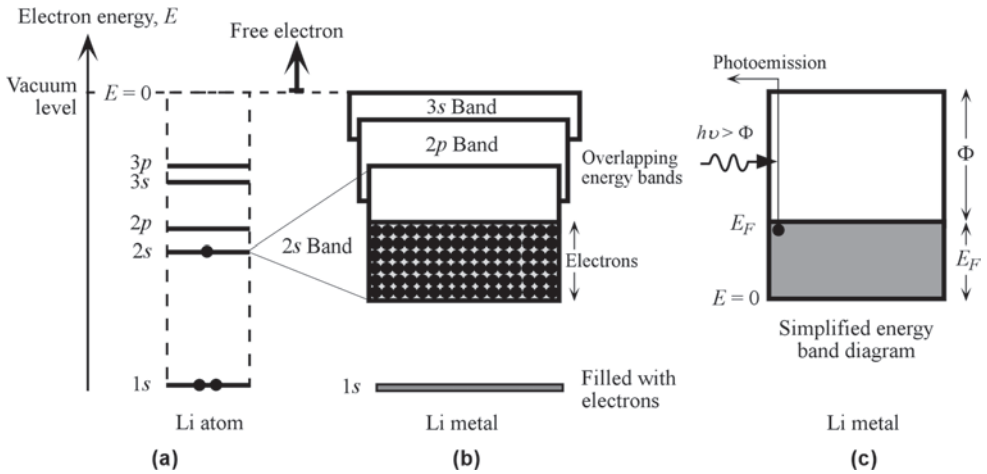


FIGURE 3.1 (a) Energy levels in a Li atom are discrete. (b) The energy levels corresponding to outer shells of isolated Li atoms form an energy band inside the crystal—for example, the 2*s* level forms a 2*s* band. Energy levels form a quasi-continuum of energy within the energy band. Various energy bands overlap to give a single band of energies that is only partially full of electrons. There are states with energies up to the vacuum level where the electron is free. (c) A simplified energy band diagram for a metal, and the photoelectric effect.

The photoemitted electrons are then accelerated by the applied field inside the multiplier tube and are made incident on other electrodes to cause secondary electron emission and hence multiplication. (One incident photon can give rise to 10^6 collected electrons.)

The electrons around E_F control many of the properties of metals such as electrical and thermal conductivity. When an electric field E is applied, an electron at E_F can gain energy from the external force eE and be able to move to a higher energy state since these are empty and available. Electrical conduction occurs by the drift of electrons at around E_F.

Many important properties of both metals (and semiconductors below) are described by considering the behavior of electrons within an energy band. In the case of metals, this is called the **free electron model**; that is, the electrons are free inside the metal and hence electron potential energy is constant. These electrons are what are normally called conduction electrons.

There are two important concepts in describing free electrons in an energy band. The **density of states** (DOS) $g(E)$ represents the number of electronic states (electron wavefunctions) in a band per unit energy per unit volume of the crystal. We can use quantum mechanics to calculate the DOS by considering how many electron wavefunctions there are within a given energy range per unit volume of the crystal. Figure 3.2 (a) shows the energy band diagram of a metal, and (b) shows, in a simplified way, how $g(E)$ depends on the electron energy in the band. According to quantum mechanics, for an electron confined within a three-dimensional potential energy well, as one might expect for a conduction electron in the crystal of a metal, DOS $g(E)$ increases with energy from the bottom of the band and is given by

Density of states in a band

$$g(E) = 4\pi (2m_e)^{3/2} h^{-3} E^{1/2} = AE^{1/2} \tag{3.1.1}$$

where E is the electron energy from the bottom of the band, m_e is the mass of the electron, h is Planck's constant, and A is a constant that represents the quantities multiplying $E^{1/2}$ in the second term. The DOS gives information on available states, that is, wavefunctions, for the electron and not on their actual occupation.

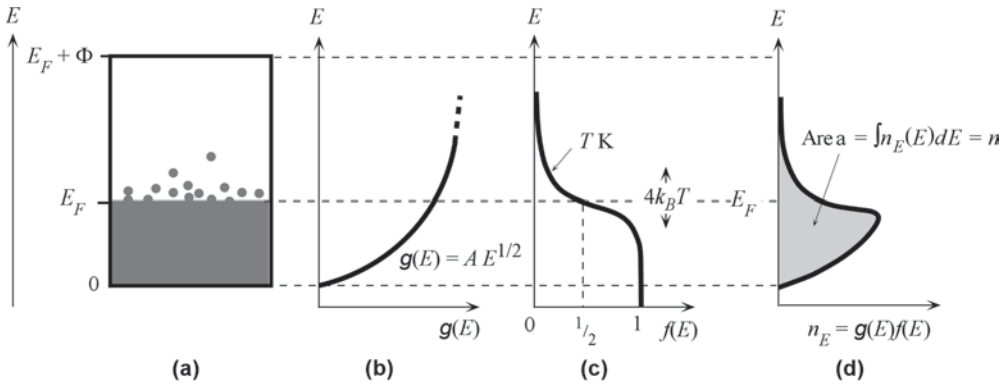


FIGURE 3.2 (a) Above 0 K, due to thermal excitation, some of the electrons are at energies above E_F. (b) The density of states, $g(E)$ vs. E in the band. (c) The probability of occupancy of a state at an energy E is $f(E)$. The product $g(E)f(E)$ is the number of electrons per unit energy per unit volume or electron concentration per unit energy. The area under the curve with the energy axis is the concentration of electrons in the band, n.

The **Fermi–Dirac function** $f(E)$ is the probability of finding an electron in a quantum state with energy E (state implies a wavefunction). It is a fundamental property of a collection of interacting electrons in *thermal equilibrium.* It is given by

$$f(E) = \frac{1}{1 + \exp\left(\dfrac{E - E_F}{k_B T}\right)} \tag{3.1.2}$$

Fermi–Dirac function

where k_B is the Boltzmann constant, T is the temperature (K), and E_F is the Fermi energy. Figure 3.2 (c) shows the dependence of $f(E)$ on the energy E. At low energies well below E_F, all states are occupied and at high energies, well above E_F, the states are empty. At E_F, $f(E_F) = 1/2$, half the states are occupied. Changes in f(E) occur around E_F over several k_BTs of energy, as indicated in Figure 3.2 (c). The higher the temperature, the greater is the reach of $f(E)$ to higher energies, and therefore more electrons at higher energies.

Consider the product $g(E)f(E)$, density of states at $E \times$ probability of finding an electron in a state at E. The product $g(E)f(E)$ represents the number electrons per unit energy per unit volume, $n_E(E)$, in the band which is shown in Figure 3.2 (d). It is essentially the energy distribution of electrons in the band. Thus, $n_E\,dE = g(E)f(E)dE$ is the number of electrons in the energy range E to $E + dE$. Integrating this from the bottom to the top of the band gives the conduction electron concentration n in the band,

$$n = \int_0^{E_F+\Phi} g(E)f(E)dE \tag{3.1.3}$$

Equations (3.1.1) and (3.1.2) can be substituted in Eq. (3.1.3) to evaluate the integration. Once integrated, the resulting equation will relate n to the Fermi energy E_F at a temperature T. Put differently, E_F depends on n, the total number of electrons per unit volume in the band. At 0 K, the Fermi energy E_{FO} is

$$E_{FO} = \left(\frac{h^2}{8m_e}\right)\left(\frac{3n}{\pi}\right)^{2/3} \tag{3.1.4}$$

Fermi energy at 0 K for metals

E_{FO} is typically a few electron volts, that is, orders of magnitude greater than k_BT. Above 0 K, E_F decreases slightly with T, though we ignore this tiny change in using metal band diagrams in optoelectronics. Inasmuch as the electrons in the metal are free, they have kinetic energies only. An electron at the Fermi energy E_{FO} has several eVs of kinetic energy, which is orders of magnitude greater than the classical kinetic theory (roughly $3k_BT/2$ for a free particle). The reason is that the electrons in the metal must obey the *Pauli exclusion principle*, which prevents more than two electrons (with spins up and down) from occupying a given energy state. Thus, the electrons must distribute themselves among available states and need to reach higher energies to satisfy the Pauli exclusion principle.

B. Energy Band Diagrams of Semiconductors

The electron energies in a semiconductor crystal, however, are distinctly different than that for metals. Figure 3.3 (a) shows a simplified two-dimensional view of the silicon crystal which has each Si atom bonding to four neighbors. All the four valence electrons per atom are used in these bonds. The interactions between the Si atoms and their valence electrons result in the electron energy in the crystal falling into two distinct energy bands called the **valence band** (VB) and **conduction band** (CB) that are separated by an energy gap, **bandgap** (E_g), as shown in Figure 3.3 (b). There are no allowed electron energies in the **bandgap**; it represents the forbidden electron energies in the crystal. The valence band represents electron wavefunctions in the crystal that correspond to bonds between the atoms. Electrons that occupy these wavefunctions are the valence electrons. Since at absolute zero of temperature all the bonds are occupied by valence electrons (there are no broken bonds), all the energy levels in the VB are normally filled with these electrons. The CB represents electron wavefunctions in the crystal that have higher energies than those in the VB and are normally empty at zero Kelvin. The top of the VB is labeled E_v, bottom of conduction band E_c, so that $E_g = E_c - E_v$ is the bandgap. The width of the CB is called the **electron affinity** χ.

An electron placed in the CB is free to move around the crystal and also to respond to an electric field because there are plenty of neighboring empty energy levels. This electron can easily gain energy from the field and move to higher energy levels because these states are

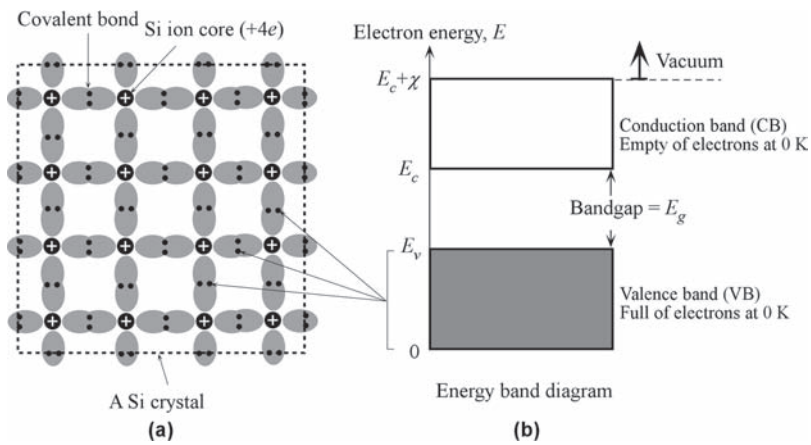


FIGURE 3.3 (a) A simplified two-dimensional view of a region of the Si crystal showing covalent bonds. (b) The energy band diagram of electrons in the Si crystal at absolute zero of temperature. The bottom of the VB has been assigned a zero of energy.

empty in the CB. Generally we can treat an electron in the CB as if it were free within the crystal by simply assigning an **effective mass** m_e^* to it. This effective mass is a quantum mechanical quantity, which takes into account that the electron in the CB interacts with a periodic potential energy as it moves through the crystal so that its inertial resistance to acceleration (definition of mass) is not the same as if it were free in vacuum.

Since the only empty states are in the CB, the excitation of an electron from the VB requires a minimum energy of E_g. Figure 3.4 illustrates what happens when an incident photon of energy $h\upsilon > E_g$ interacts with an electron in the VB. This electron absorbs the incident photon and gains sufficient energy to surmount the energy gap E_g and reach the CB. Consequently, a free electron in the CB and a "hole," corresponding to a missing electron in the VB, are created. In some semiconductors, such as Si and Ge, the photon absorption process also involves lattice vibrations (vibrations of the Si atoms) which we have not shown in Figure 3.4.

The empty electronic state, or the missing electron in the bond, is what we call a **hole** in the valence band. The free electron, which is in the CB, can wander around the crystal and contribute to the electrical conduction when an electric field is applied. The region remaining around the hole in the VB is positively charged because a charge of $-e$ has been removed from an otherwise neutral region of the crystal. This hole, denoted as h^+, can also wander around the crystal as if it were "free." This is because an electron in a neighboring bond can "jump," that is, tunnel into the hole, to fill the vacant electronic state at this site and thereby create a hole at its original position as illustrated in Figure 3.5. This is effectively equivalent to the hole being displaced in the opposite direction. Thus, conduction in semiconductors occurs by both electrons and holes with charges $-e$ and $+e$ and their own effective masses m_e^* and m_h^*.

When a wandering electron in the CB meets a hole in the VB, it has found an empty electronic state of lower energy and it therefore occupies it. The electron falls from the CB to the VB to fill the hole. This is called **recombination**, which results in the annihilation of an electron in the CB and a hole in the VB as in Figure 3.5. The excess energy of the electron falling from CB to VB in certain semiconductors such as GaAs and InP is emitted as a photon. In Si and Ge the excess energy is lost as lattice vibrations (heat), that is, as phonons. In the steady state, the thermal generation rate is balanced by the recombination rate so that the electron concentration n in the CB and hole concentration p in the VB remain constant; both n and p depend on the temperature.

Although in the specific example in Figure 3.4, a photon of energy $h\upsilon > E_g$ creates an electron–hole pair, other sources of energy can also lead to an electron–hole pair creation. In fact, in the absence of radiation, there is still an electron and hole pair generation process going on in the sample as a result of **thermal generation**. Due to thermal energy, the atoms in the crystal

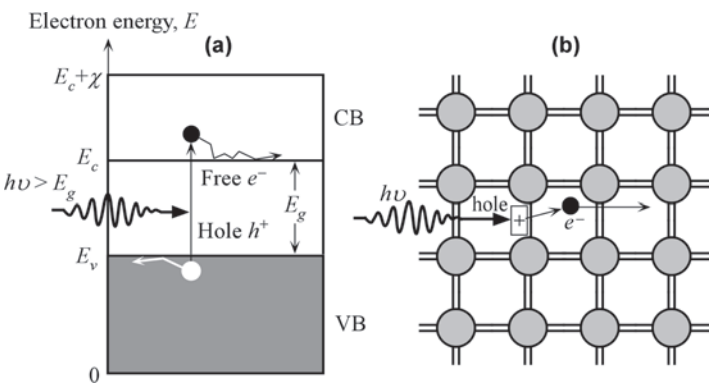


FIGURE 3.4 (a) A photon with an energy $h\upsilon$ greater than E_g can excite an electron from the VB to the CB. (b) Each line with Si-Si atoms is a valence electron in a bond. When a photon breaks a Si-Si bond, a free electron and a hole in the Si-Si bond are created. The result is the photogeneration of an electron–hole pair (EHP).

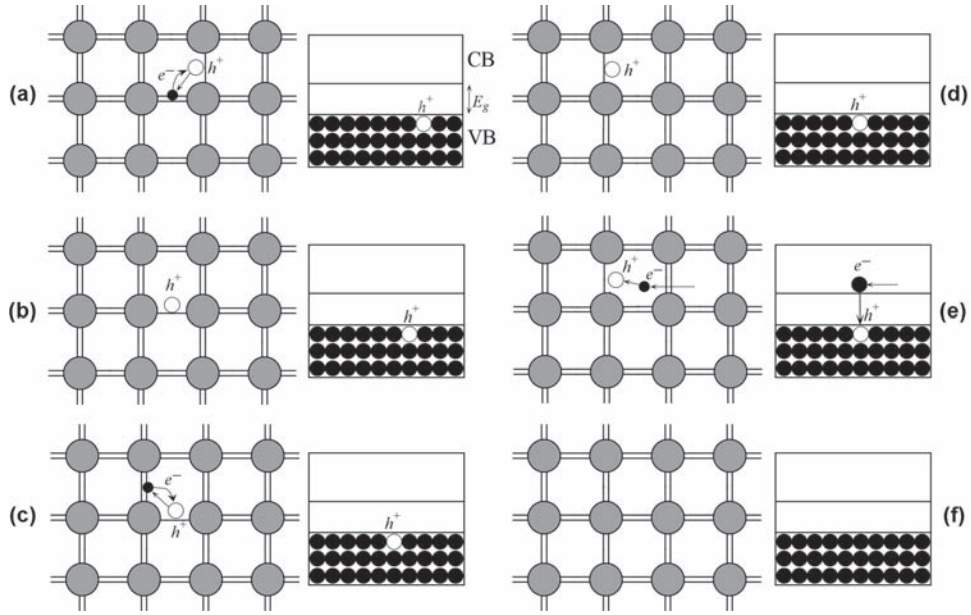


FIGURE 3.5 A pictorial illustration of a hole in the valence band wandering around the crystal due to the tunneling of electrons from neighboring bonds, and its eventual recombination with a wandering electron in the conduction band. A missing electron in a bond represents a hole as in (a). An electron in a neighboring bond can tunnel into this empty state and thereby cause the hole to be displaced as in (a) to (d). The hole is able to wander around in the crystal as if it were free but with a different effective mass than the electron. A wandering electron in the CB meets a hole in the VB in (e), which results in the recombination and the filling of the empty VB state as in (f).

are constantly vibrating which corresponds to the bonds between the Si atoms being periodically deformed with a distribution of energies. Energetic vibrations can rupture bonds and thereby create electron–hole pairs, which correspond to the excitation of electrons from the VB to the CB.

3.2 SEMICONDUCTOR STATISTICS

Many important properties of semiconductors are described by considering the behavior of electrons in the CB and holes in the VB. We need to consider the density of states (DOS) $g_{CB}(E)$ of electronic states in the CB, $g_{VB}(E)$ in the VB and their occupation statistics. Figure 3.6 (a) shows the energy band diagram of a semiconductor, and (b) shows, in a simplified way, how $g_{CB}(E)$ depends on the electron energy in the CB and VB near the band edges E_c and E_v, respectively. The DOS $g_{CB}(E)$ increases with energy from the CB edge as

Density of states near E_c

$$g_{CB}(E) = 4\pi(2m_e^*)^{3/2}h^{-3}(E - E_c)^{1/2} \quad \textbf{(3.2.1)}$$

where $(E - E_c)$ is the electron energy from the bottom of the CB. The DOS curves usually have various peaks at certain energies for crystalline solids, which we have neglected as these are not needed for the present discussion.

To find the energy distribution of electrons in the CB, we need the Fermi–Dirac function $f(E)$, the probability of finding an electron in a quantum state with energy E. The behavior of $f(E)$ is shown in Figure 3.6 (c) assuming that the Fermi level E_F is located in the bandgap. (Its position

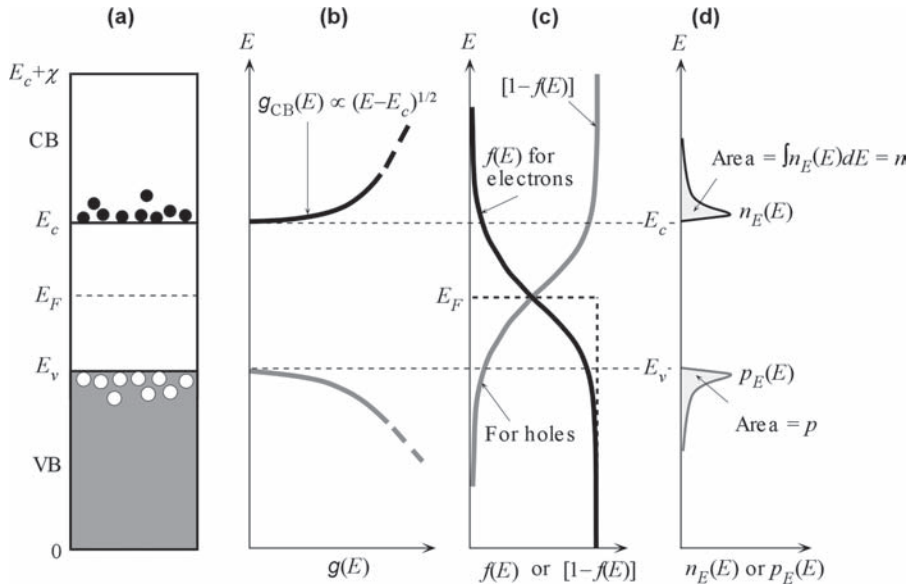


FIGURE 3.6 (a) Energy band diagram of a semiconductor, assuming that the Fermi level is near the middle of the bandgap. (b) Density of states (number of states per unit energy per unit volume). (c) Fermi–Dirac probability function (probability of occupancy of a state). (d) The product of $g(E)$ and $f(E)$ is the energy density of electrons in the CB (number of electrons per unit energy per unit volume). The area under $n_E(E)$ vs. E is the electron concentration.

in the energy gap is discussed later.) When we multiply $g(E)$ with $f(E)$, we get the number of electrons per unit energy per unit volume at E in the CB, $n_E(E)$, which is shown in Figure 3.6 (d). Thus, $n_E dE = g_{CB}(E)f(E)dE$ is the number of electrons in the energy range [Figure 3.6 (d)] E to $E + dE$. Integrating this from the bottom (E_c) to the top ($E_c + \chi$) of the CB gives the electron concentration

$$n = \int_{E_c}^{E_c+\chi} g_{CB}(E)\, f(E)\, dE \qquad \textbf{(3.2.2)}$$

Electron concentration in CB

Whenever $(E_c - E_F) \gg k_B T$, that is, E_F is at least a few $k_B T$ below E_c, then $f(E) \approx \exp\left[-(E - E_F)/k_B T\right]$. That is, the Fermi–Dirac statistics can be replaced by *Boltzmann* statistics. Such semiconductors are called **nondegenerate**. It implies that the number of electrons in the CB is far less than the number states in this band. For nondegenerate semiconductors, the above integration leads to

$$n = N_c \exp\left(-\frac{E_c - E_F}{k_B T}\right) \qquad \textbf{(3.2.3)}$$

Electron concentration in CB

where $N_c = 2\left[2\pi m_e^* k_B T/h^2\right]^{3/2}$ is a constant at a given temperature for a given material, called the **effective density of states at the CB edge**. The result of the integration in Eq. (3.2.3) seems too simple but it is, however, an approximation as it assumes that $(E_c - E_F) \gg k_B T$. We can interpret Eq. (3.2.3) as follows. If we take all the states in the conduction band and replace them with an effective concentration N_c (number of states per unit volume) at E_c and then multiply this simply by the Boltzmann probability function, $f(E_c) = \exp\left[-(E_c - E_F)/k_B T\right]$, we would

obtain the concentration of electrons at E_c, that is, in the conduction band. N_c is thus an effective density of states at the CB band edge.

We can carry out a similar analysis for the concentration of holes in the VB as illustrated in Figures 3.6 (a)–(d). Multiplying the density of states $g_{VB}(E)$ in the VB with the probability of occupancy by a hole, $\left[1 - f(E)\right]$, gives p_E, the hole concentration per unit energy. Note that the probability of finding a hole, a missing electron, in a state with energy E is $1 - f(E)$. Integrating $p_E(E)$ over the VB gives the hole concentration by assuming that E_F is a few k_BT above E_v, we obtain

Hole concentration in VB

$$p \approx N_v \exp\left(-\frac{E_F - E_v}{k_BT}\right) \tag{3.2.4}$$

where $N_v = 2\left[2\pi m_h^* k_B T/h^2\right]^{3/2}$ is the **effective density of states at the VB edge**, and m_h^* is the effective mass of holes in the VB.

There are no specific assumptions in our derivations above, except for E_F being a few k_BT away from the band edges, which means that Eqs. (3.2.3) and (3.2.4) are generally valid. It is apparent from Eqs. (3.2.3) and (3.2.4) that the location of E_F determines the electron and hole concentrations. Thus, E_F can be viewed as a useful semiconductor property that allows us to represent electron and hole concentrations in the energy band diagram. In an intrinsic semiconductor (a pure crystal), $n = p$, by using Eqs. (3.2.3) and (3.2.4) we can show that the Fermi level E_{Fi} in the *intrinsic crystal* is above E_v and located in the bandgap at

Fermi level in intrinsic crystal

$$E_{Fi} = E_v + \tfrac{1}{2} E_g - \tfrac{1}{2} kT \ln\left(\frac{N_c}{N_v}\right) \tag{3.2.5}$$

Typically N_c and N_v values are comparable and both occur in the logarithmic term so that E_{Fi} is approximately in the middle of the bandgap as originally sketched in Figure 3.6.

There is a useful semiconductor relation between n and p, called the **mass action law**. From Eqs. (3.2.3) and (3.2.4), the product np is

Mass action law

$$np = N_c N_v \exp\left(-\frac{E_g}{k_BT}\right) = n_i^2 \tag{3.2.6}$$

where $E_g = E_c - E_v$ is the bandgap energy and n_i^2 has been *defined as* $N_c N_v \exp(-E_g/k_BT)$. The quantity n_i is a material property; for example, it depends on E_g, but not on the position of the Fermi level. It also depends on the temperature. The intrinsic concentration n_i corresponds to the concentration of electrons or holes in an undoped (pure) crystal, that is, intrinsic semiconductor. In such a semiconductor $n = p = n_i$, which is therefore called the **intrinsic concentration**. The mass action law is valid whenever we have thermal equilibrium and the sample is in the dark (without any photogeneration).

Equations (3.2.3) and (3.2.4) determine the total concentration of electrons and holes in the CB and VB, respectively. The average energy of the electrons in the CB can be calculated by using $n_E(E)$, their energy distribution. The result gives the average energy as $(3/2)k_BT$ above E_c. Since the electron in the CB is "free" in the crystal with an effective mass m_e^*, it wanders around the crystal with an average kinetic energy $(3/2)k_BT$, which is the same as the kinetic energy of a free atom in a gas or vapor in a tank. This is not surprising as both particles are wandering freely (without interacting with each other) and obey Boltzmann statistics. If v is the electron velocity and triangular brackets represent an average, then $<(1/2)m_e^* v^2>$ must be $(3/2)k_BT$. We can thus calculate the *root mean square velocity* $\left(\langle v^2 \rangle\right)^{1/2}$ which is called the **thermal velocity** (v_{th}) and is typically $\sim 10^5$ m s^{-1}. The same ideas apply to holes in the VB with an effective hole mass m_h^*.

It is important to understand what the **Fermi energy** represents. First, it is a convenient way to represent free carrier concentrations (n in the CB and p in the VB) on the energy band diagram. However, the most useful property of E_F is in terms of a change in E_F. Any change ΔE_F across a material system represents electrical work input or output per electron.[2] If V is the potential difference between two points in a material system, then

Fermi energy and electrical work

$$\Delta E_F = eV \tag{3.2.7}$$

For a semiconductor system in equilibrium, in the dark, and with no applied voltage or no EMF generated, $\Delta E_F = 0$ and E_F must be uniform across the system.

3.3 EXTRINSIC SEMICONDUCTORS

A. *n*-Type and *p*-Type Semiconductors

By introducing small amounts of impurities into an otherwise-pure crystal, it is possible to obtain a semiconductor in which the concentration of carriers of one polarity is much in excess of the other type. Such semiconductors are referred to as **extrinsic semiconductors** vis-à-vis the intrinsic case of a pure and perfect crystal. For example, by adding pentavalent impurities, such as arsenic, which have a valency one more than Si, we can obtain a semiconductor in which the electron concentration is much larger than the hole concentration. In this case we will have an ***n*-type semiconductor**. If we add trivalent impurities, such as boron, which have a valency of one less than four, we then have an excess of holes over electrons—a ***p*-type semiconductor**.

An arsenic (As) atom has five valence electrons whereas Si has four. When the Si crystal is doped with small amounts of As, each As atom substitutes for one Si atom and is surrounded by four Si atoms. When an As atom bonds with four Si atoms, it has one electron left unbounded. This fifth electron cannot find a bond to go into so it is left orbiting around the As atom, which looks like an As^+, as illustrated in Figure 3.7 (a). The As^+ ionic center, with an electron e^- orbiting

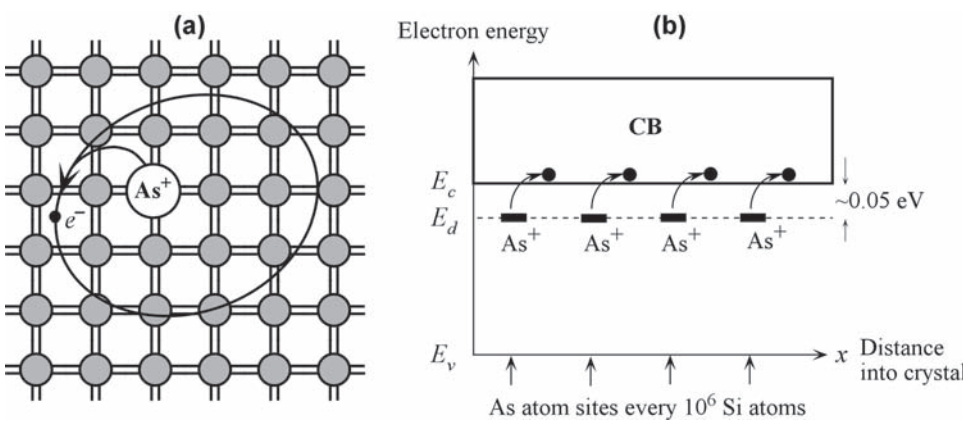


FIGURE 3.7 (a) The four valence electrons of As allow it to bond just like Si but the fifth electron is left orbiting the As site. The energy required to release to free fifth electron into the CB is very small. (b) Energy band diagram for an n-type Si doped with 1 ppm As. There are donor energy levels just below E_c around As^+ sites.

[2]For readers familiar with thermodynamics, its rigorous definition is that E_F is the *chemical potential per electron*, that is, Gibbs free energy per electron. The definition in Eqs. (3.2.7) is in terms of a change in E_F.

it, resembles a hydrogen atom in a silicon environment. We can easily calculate how much energy is required to free this electron away from the As site, thereby ionizing the As impurity by using our knowledge on the ionization of a hydrogen atom (removing the electron from the H-atom). This energy turns out to be a few hundredths of an electronvolt, that is, ~0.05 eV, which is comparable to the thermal energy at room temperature ($\sim k_B T = 0.025$ eV). Thus, the fifth valence electron can be readily freed by thermal vibrations of the Si lattice. The electron will then be "free" in the semiconductor, or in other words, it will be in the CB. The energy required to excite the electron to the CB is therefore ~0.05 eV. The addition of As atoms introduces localized electronic states at the As sites because the fifth electron has a localized wavefunction, of the hydrogenic type, around As^+. The energy of these states, E_d, is ~0.05 eV below E_c because this is how much energy is required to take the electron away into the CB. Thermal excitation by lattice vibrations at room temperature is sufficient to ionize the As atom, that is, excite the electron from E_d into the CB. This process creates free electrons; however, the As^+ ions remain immobile as shown in the energy band diagram of an *n*-type semiconductor in Figure 3.7 (b).

Because the As atom donates an electron into the CB, it is called a **donor** impurity. E_d is the electron energy around the donor atom and it is below E_c by ~0.05 eV as in Figure 3.7 (b). If N_d is the donor atom concentration in the crystal, provided that $N_d \gg n_i$, then at room temperature, the electron concentration in the CB will nearly be equal to N_d, that is, $n \approx N_d$. The hole concentration will be $p = n_i^2/N_d$, which is less than the intrinsic concentration. The reason is that a small fraction of the large number of electrons in the CB recombine with holes in the VB so as to maintain $np = n_i^2$. The mass action law must be maintained in equilibrium.

The conductivity σ of a semiconductor depends on both electrons and holes as both contribute to charge transport. If μ_e and μ_h are the drift mobilities of the electrons and holes, respectively, then

Semiconductor conductivity

$$\sigma = en\mu_e + ep\mu_h \tag{3.3.1}$$

which for an *n*-type semiconductor becomes,

n-type conductivity

$$\sigma = eN_d\mu_e + e\left(\frac{n_i^2}{N_d}\right)\mu_h \approx eN_d\mu_e \tag{3.3.2}$$

We should, by similar arguments to the above, anticipate that doping a Si crystal with a trivalent atom (valency of 3) such as B (boron) will result in a *p*-type Si that has an excess of holes in the crystal. Consider doping Si with small amounts of B as shown in Figure 3.8 (a). Because B has only three valence electrons, when it shares them with four neighboring Si atoms one of the bonds has a missing electron which is of course a "hole." A nearby electron can tunnel into this hole and displace the hole further away from the B atom. As the hole moves away it gets attracted by the negative charge left behind on the B atom. The binding energy of this hole to the B^- ion (a B atom that has accepted an electron) can be calculated using the hydrogenic atom analogy just like in the *n*-type Si case. This binding energy also turns out to be very small, ~0.05 eV, so that at room temperature the thermal vibrations of the lattice can free the hole away from the B^- site. A free hole, we recall, exists in the VB. The escape of the hole from the B^- site involves the B atom accepting an electron from a neighboring Si-Si bond (from the VB) which effectively results in the hole being displaced away, and its eventual escape to freedom in the VB. The B atom introduced into the Si crystal therefore acts as an electron **acceptor** impurity. The electron accepted by the B atom comes from a nearby bond. On the energy band diagram, an electron leaves the VB and gets accepted by a B atom which becomes negatively charged. This process leaves a hole in the VB which is free to wander away as illustrated in Figure 3.8 (b).

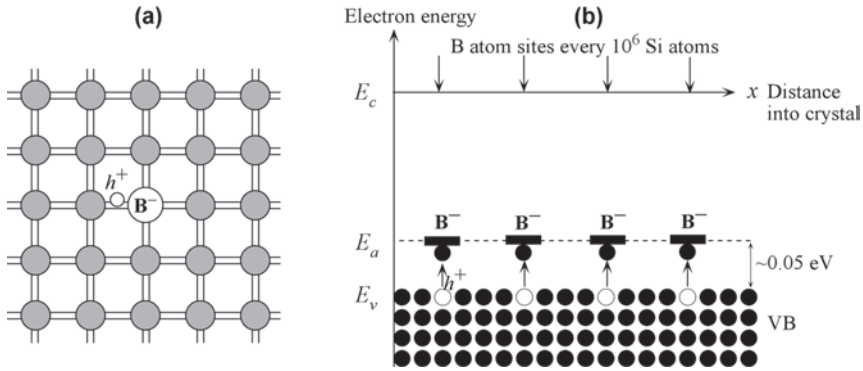


FIGURE 3.8 (a) Boron-doped Si crystal. B has only three valence electrons. When it substitutes for a Si atom one of its bonds has an electron missing and therefore a hole. (b) Energy band diagram for a *p*-type Si doped with 1 ppm B. There are acceptor energy levels just above E_v around B^- sites. These acceptor levels accept electrons from the VB and therefore create holes in the VB.

It is apparent that doping a silicon crystal with a trivalent impurity results in a *p*-type material. We have many more holes than electrons for electrical conduction since the negatively charged B atoms are immobile and hence cannot contribute to the conductivity. If the concentration of acceptor impurities N_a in the crystal is much greater than the intrinsic concentration n_i, then at room temperature all the acceptors would have been ionized and thus $p \approx N_a$. The electron concentration is then determined by the mass action law, $n = n_i^2/N_a$, which is much smaller than p, and consequently the conductivity is simply given by $\sigma = eN_a\mu_h$, similar to Eq. (3.3.2).

Figures 3.9 (a)–(c) show the energy band diagrams of an intrinsic, an *n*-type, and a *p*-type semiconductor. The energy distance of E_F from E_c and E_v determines the electron and hole concentrations by virtue of Eqs. (3.2.3) and (3.2.4). Note the locations of the Fermi level in each case: E_{Fi} for intrinsic; E_{Fn} for *n*-type; and E_{Fp} for *p*-type.

The position of the Fermi level in the energy band diagram plays an important role in understanding device principles. It is clear from Eqs. (3.2.3) and (3.2.4) that its location in the energy band diagram acts as a lever in finding the free electron and hole concentrations in the CB and VB, respectively, as highlighted in Figures 3.9 (a)–(c). It represents the electron

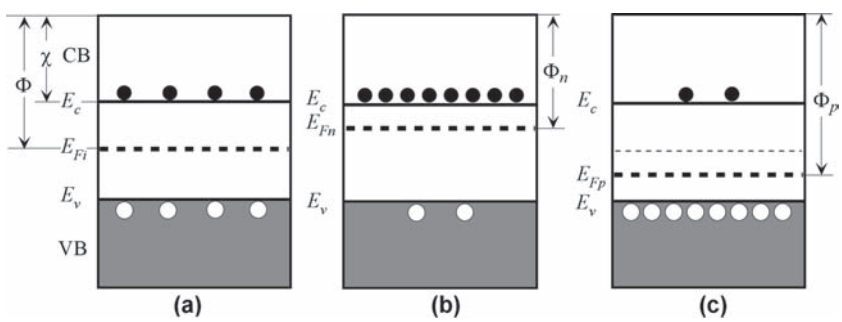


FIGURE 3.9 Energy band diagrams for (a) intrinsic, (b) *n*-type, and (c) *p*-type semiconductors. In all cases, $np = n_i^2$. Note that donor and acceptor energy levels are not shown. CB = conduction band, VB = valence band, E_c = CB edge, E_v = VB edge, E_{Fi} = Fermi level in intrinsic semiconductor, E_{Fn} = Fermi level in *n*-type semiconductor, E_{Fp} = Fermi level in *p*-type semiconductor. χ is the electron affinity. Φ, Φ_n, and Φ_p are the work functions for the intrinsic, *n*-type, and *p*-type semiconductors, respectively.

TABLE 3.1 Selected typical properties of various semiconductors at 300 K

	a (nm)	E_g (eV)	χ (eV)	$N_c(\text{cm}^{-3})$	$N_v(\text{cm}^{-3})$	$n_i(\text{cm}^{-3})$	ε_r	μ_e $(\text{cm}^2\text{V}^{-1}\text{s}^{-1})$	μ_h $(\text{cm}^2\text{V}^{-1}\text{s}^{-1})$
Ge (DI)	0.5650	0.66 (I)	4.13	1.04×10^{19}	6.0×10^{19}	2.3×10^{13}	16.0	3900	1900
Si (DI)	0.5431	1.11 (I)	4.05	2.8×10^{19}	1.2×10^{19}	1.0×10^{10}	11.8	1450	490
InP (ZB)	0.5868	1.35 (D)	4.50	5.2×10^{17}	1.1×10^{19}	3.0×10^{7}	12.6	4600	150
GaAs (ZB)	0.5653	1.42 (D)	4.07	4.7×10^{17}	7.0×10^{18}	2.1×10^{6}	13.0	8500	400
AlAs (ZB)	0.5661	2.17 (I)	3.50	1.5×10^{19}	1.7×10^{19}	10	10.1	200	100

Notes: Data combined from a number of sources. I and D represent indirect and direct bandgap. DI, diamond crystal; ZB, zinc blend; a, lattice constant. (Note that there are variations in the values of certain properties among books, for example, n_i for Si, ε_r for GaAs, *etc.* Most commonly used or recent values have been selected.)

and hole concentrations in a semiconductor in the energy band diagram. The work function Φ of a semiconductor is defined as the energy distance from the Fermi level to the vacuum level as shown in Figures 3.9 (a)–(c) as Φ, Φ_n, and Φ_p for the intrinsic, n-type, and p-type semiconductor, respectively. Although there are no states at E_F, the work function nonetheless represents the *average energy* required to remove an electron from a semiconductor.[3]

The following definitions and notation are generally used to describe extrinsic semiconductors. Electrons in an n-type semiconductor ($n > p$) are **majority carriers** whereas holes are **minority carriers**. The electron, that is, majority carrier, concentration in this n-type semiconductor in equilibrium is n_{no}, where the subscripts refer to the n-type semiconductor and equilibrium (which excludes photoexcitation). The hole, that is, minority carrier, concentration is denoted p_{no}. In this notation, $n_{no} = N_d$ and the mass action law is $n_{no}p_{no} = n_i^2$. Similarly, hole or majority carrier concentration in a p-type semiconductor ($p > n$) is p_{po} and electron, minority carrier, concentration is n_{po}. Further $p_{po} = N_a$ and $n_{po}p_{po} = n_i^2$.

Table 3.1 summarizes some of the properties of useful semiconductors, including Si and GaAs, at 300 K.

B. Compensation Doping

Compensation doping describes the doping of a semiconductor with both donors and acceptors to control the properties. For example, a p-type semiconductor doped with N_a acceptors can be converted to an n-type semiconductor by simply adding donors until the concentration N_d exceeds N_a. The effect of donors compensates for the effect of acceptors and vice versa. The electron concentration is then given by $N_d - N_a$ provided the latter is larger than n_i. When both acceptors and donors are present, what essentially happens is that electrons from donors recombine with the holes from the acceptors so that the mass action law $np = n_i^2$ is obeyed. Remember that we cannot simultaneously increase the electron and hole concentrations because that leads to an increase in the recombination rate which returns the electron and hole concentrations to values that satisfy $np = n_i^2$. Suppose that we have more donors than acceptors. When an acceptor atom accepts a valence band electron, a hole is created in the VB. This hole then recombines with an electron from the CB; the electron would have been supplied by a donor. If we take the initial electron concentration as $n = N_d$ then the recombination between the electrons from the donors and N_a holes generated by N_a acceptors

[3]The following intuitive argument may help. The easiest electron to remove from an intrinsic semiconductor needs an energy χ. To maintain equilibrium, we must also remove an electron from the VB, which needs an energy $E_g + \chi$ so that the average energy is $\frac{1}{2}E_g + \chi$, which is the work function of the intrinsic semiconductor.

results in the electron concentration reduced by N_a to $n = N_d - N_a$. The hole concentration is then $p = n_i^2/(N_d - N_a)$. By a similar argument, if we have more acceptors than donors, the hole concentration becomes $p = N_a - N_d$, with electrons from N_d donors recombining with holes from N_a acceptors. The electron concentration is then $n = n_i^2/(N_a - N_d)$.

C. Nondegenerate and Degenerate Semiconductors

In nondegenerate semiconductors, the number of states in the CB far exceeds the number of electrons so that the likelihood of two electrons trying to occupy the same state is almost nil. This means that the Pauli exclusion principle can be neglected and the electron statistics can be described by the Boltzmann statistics. N_c is a measure of the density of states in the CB. The Boltzmann expression in Eq. (3.2.3) for n is valid only when $n \ll N_c$. Those semiconductors for which $n \ll N_c$ and $p \ll N_v$ are termed **nondegenerate** semiconductors.

When the semiconductor has been excessively doped with donors, n may be so large, typically $10^{19}-10^{20}\,\text{cm}^{-3}$, that it may be comparable to N_c. In that case the Pauli exclusion principle becomes important in the electron statistics and we have to use the Fermi–Dirac statistics. Such a semiconductor exhibits properties that are more metal-like than semiconductor. Semiconductors that have $n \gtrsim N_c$ or $p \gtrsim N_v$ are called **degenerate semiconductors**. For degenerate semiconductors, the Fermi level is inside the band that is involved in electrical conduction. For example, the Fermi level in an n-type degenerate semiconductor is above E_c.

The large carrier concentration in a degenerate semiconductor is due to the heavy doping. For example, as the donor concentration in an n-type semiconductor is increased, at sufficiently high doping levels, the donor atoms become so close to each other that their orbitals overlap to form a narrow energy band, which overlaps and becomes part of the conduction band as illustrated in Figure 3.10 (a). This situation is reminiscent to the valence electrons filling overlapping energy bands in a metal. In a degenerate n-type semiconductor, the Fermi level is therefore within the CB, or above E_c just like E_F is within the band in a metal. Majority of the states between E_c and E_F are full of electrons as indicated in Figure 3.10 (a). Notice that E_c has been pushed down so that the bandgap is *narrower* in heavily doped semiconductors. The random distribution of a large number of donors in the crystal also introduces a random variation in the potential energy (PE) of the electron; the PE is no longer perfectly periodic. Put differently, there are spatial variations in the density of states in the crystal. Such random fluctuations in the PE give rise to a small tail (an extension) in the density of states $g(E)$ in the bandgap, called **band tailing**, so that E_c is no longer sharp in heavily doped crystals.

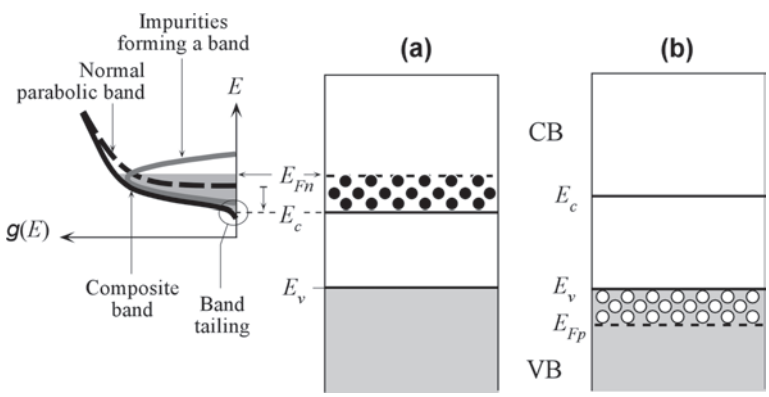


FIGURE 3.10
(a) Degenerate n-type semiconductor. Large number of donors form a band that overlaps the CB. E_c is pushed down and E_{Fn} is within the CB. (b) Degenerate p-type semiconductor.

In the case of a p-type degenerate semiconductor, the Fermi level lies in the VB below E_v as in Figure 3.10 (b). One cannot simply assume that $n = N_d$ or $p = N_a$ in a degenerate semiconductor because the dopant concentration is so large that they interact with each other. Not all dopants are able to become ionized and the carrier concentration eventually reaches a saturation typically around ~10^{20} cm^{-3}. Furthermore, the mass action law, $np = n_i^2$, is not valid for degenerate semiconductors.

D. Energy Band Diagrams in an Applied Field

Consider an n-type semiconductor that is connected to a voltage supply of V and is carrying a current as shown in Figure 3.11. From basic electrical conduction we know that there is an electric field E inside the semiconductor that drifts the conduction electrons and hence causes a current flow. Consider now the energy band diagram. The Fermi level E_F is above that for the intrinsic case (E_{Fi}), closer to E_c than E_v. The applied voltage drops uniformly along the semiconductor so that the electrons in the semiconductor now also have an imposed electrostatic potential energy, which decreases toward the positive terminal as illustrated in Figure 3.11. The whole band structure, the CB and the VB, therefore tilts. When an electron drifts from A toward B, its PE decreases because it is approaching the positive terminal.

For a semiconductor system in the dark, in equilibrium and with no applied voltage or no EMF generated, E_F must be uniform across the system since $\Delta E_F = eV = 0$. However, when electrical work is done on the system, for example, when a battery is connected to a semiconductor, E_F is not uniform throughout the whole system. A change ΔE_F in E_F within a material system is equivalent to electrical work per electron or eV. The Fermi level E_F therefore follows the electrostatic PE behavior. The change in E_F from one end to the other, $E_F(A) - E_F(B)$, is just eV, the energy needed in taking an electron through the semiconductor as shown in Figure 3.11. The electron concentration in the semiconductor is uniform so that $E_c - E_F$ must be constant from one end to the other. Thus the CB, VB, and E_F all bend by the same amount.

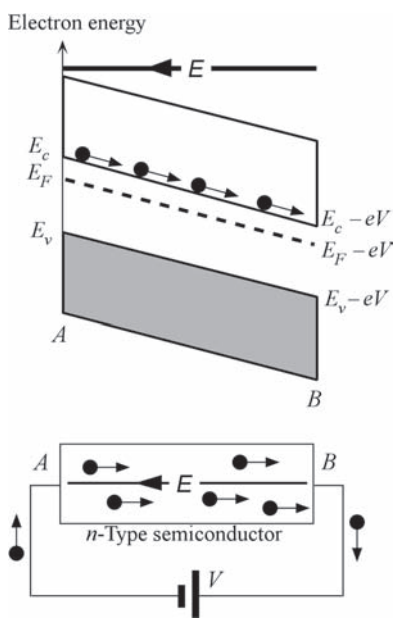


FIGURE 3.11 Energy band diagram of an n-type semiconductor connected to a voltage supply of V volts. The whole energy diagram tilts because the electron now has an electrostatic potential energy as well.

EXAMPLE 3.3.1 Fermi levels in semiconductors

An *n*-type Si wafer has been doped uniformly with 10^{16} phosphorus (P) atoms cm^{-3}. Calculate the position of the Fermi energy with respect to the Fermi energy E_{Fi} in intrinsic Si. The above *n*-type Si sample is further doped with 2×10^{17} boron atoms cm^{-3}. Calculate the position of the Fermi energy with respect to the Fermi energy E_{Fi} in intrinsic Si at room temperature (300 K), and hence with respect to the Fermi energy in the *n*-type case above.

Solution

P (Group V) gives *n*-type doping with $N_d = 10^{16}\,cm^{-3}$, and since $N_d \gg n_i\,(=10^{10}\,cm^{-3}$ from Table 3.1), we have $n = N_d = 10^{16}\,cm^{-3}$. For intrinsic Si

$$n_i = N_c \exp\left[-(E_c - E_{Fi})/k_B T\right]$$

whereas for doped Si,

$$n = N_c \exp\left[-(E_c - E_{Fn})/k_B T\right] = N_d$$

where E_{Fi} and E_{Fn} are the Fermi energies in the intrinsic and *n*-type Si. Dividing the two expressions,

$$N_d/n_i = \exp\left[(E_{Fn} - E_{Fi})/k_B T\right]$$

so that

$$E_{Fn} - E_{Fi} = k_B T \ln(N_d/n_i) = (0.0259\,\text{eV}) \ln(10^{16}/10^{10}) = 0.358\,\text{eV}$$

When the wafer is further doped with boron, the acceptor concentration, $N_a = 2 \times 10^{17}\,cm^{-3} > N_d = 10^{16}\,cm^{-3}$. The semiconductor is compensation doped and compensation converts the semiconductor to a *p*-type Si. $p = N_a - N_d = 2 \times 10^{17} - 10^{16} = 1.9 \times 10^{17}\,cm^{-3}$.

For intrinsic Si

$$p = n_i = N_v \exp\left[-(E_{Fi} - E_v)/k_B T\right],$$

whereas for doped Si

$$p = N_v \exp\left[-(E_{Fp} - E_v)/k_B T\right] = N_a - N_d$$

where E_{Fi} and E_{Fp} are the Fermi energies in the intrinsic and *p*-type Si, respectively. Dividing the two expressions

$$p/n_i = \exp\left[-(E_{Fp} - E_{Fi})/k_B T\right]$$

so that

$$E_{Fp} - E_{Fi} = -k_B T \ln(p/n_i) = -(0.0259\,\text{eV}) \ln(1.9 \times 10^{17}/1.0 \times 10^{10}) = -0.434\,\text{eV}$$

The negative value indicates that E_{Fp} is below E_{Fi}.

EXAMPLE 3.3.2 Conductivity of *n*-Si

Consider a pure intrinsic Si crystal. What would be its intrinsic conductivity at 300 K? What are the electron and hole concentrations in an *n*-type Si crystal that has been doped with $10^{16}\,cm^{-3}$ phosphorus (P) donors. What is the conductivity if the drift mobility of electrons is about 1200 $cm^2\,V^{-1}\,s^{-1}$ at this concentration of dopants?

Solution

The intrinsic concentration $n_i = 1 \times 10^{10}\,\text{cm}^{-3}$, so that the intrinsic conductivity is

$$\sigma = en_i(\mu_e + \mu_h) = (1.6 \times 10^{10}\,\text{C})(1 \times 10^{10}\,\text{cm}^{-3})(1450 + 490\,\text{cm}^2\,\text{V}^{-1}\,\text{s}^{-1})$$

$$= 3.1 \times 10^{-6}\,\Omega^{-1}\,\text{cm}^{-1} \quad \text{or} \quad 3.1 \times 10^{-4}\,\Omega^{-1}\,\text{m}^{-1}$$

Consider *n*-type Si. $N_d = 10^{16}\,\text{cm}^{-3} > n_i(=10^{10}\,\text{cm}^{-3})$, the electron concentration $n = N_d = 10^{16}\,\text{cm}^{-3}$ and $p = n_i^2/N_d = (10^{10}\,\text{cm}^{-3})^2/(10^{16}\,\text{cm}^{-3}) = 10^4\,\text{cm}^{-3}$; and negligible compared to *n*.
The conductivity is

$$\sigma = eN_d\mu_e = (1.6 \times 10^{-19}\,\text{C})(1 \times 10^{16}\,\text{cm}^{-3})(1200\,\text{cm}^2\,\text{V}^{-1}\,\text{s}^{-1}) = 1.92\,\Omega^{-1}\,\text{cm}^{-1}$$

3.4 DIRECT AND INDIRECT BANDGAP SEMICONDUCTORS: *E–k* DIAGRAMS

We know from quantum mechanics that when the electron is within an infinite potential energy well of spatial width *L*, its energy is quantized and given by

$$E_n = \frac{(\hbar k_n)^2}{2m_e}$$

where m_e is the mass of the electron and the electron's wave vector k_n is essentially a quantum number determined by

$$k_n = \frac{n\pi}{L}$$

where $n = 1, 2, 3, \ldots$. The energy increases parabolically with the wave vector k_n. We also know that the electron momentum is given by $\hbar k_n$. This description can be used to represent the behavior of electrons in a metal within which their average potential energy can roughly be taken to be zero. In other words, we take $V(x) = 0$ within the metal crystal and $V(x)$ to be large, for example, $V(x) = V_o$ (several electron volts) outside, so that the electron is contained within the metal. This is the **nearly free electron model** of a metal which has been quite successful in interpreting many of the metallic properties. Indeed, we typically calculate the density of states $g(E)$ based on the three-dimensional potential well problem. However, it is quite obvious that this model is too simple because it does not take into account the actual variation of the electron potential energy in the crystal.

The potential energy of the electron depends on its location within the crystal and is periodic due to the regular arrangement of the atoms. How does a periodic potential energy affect the relationship between *E* and *k*? It will no longer be simply $E_n = (\hbar k_n)^2/2m_e$.

To find the energy of the electron in a crystal, we need to solve the Schrödinger equation for a periodic potential energy function in three dimensions. We first consider the hypothetical one-dimensional crystal shown in Figure 3.12. The electron potential energy functions for each atom add to give an overall potential energy function $V(x)$, which is clearly periodic in *x* with the periodicity of the crystal, *a*. Thus, $V(x) = V(x + a) = V(x + 2a) = \ldots$ and so on. Our task is therefore to solve the Schrödinger equation for the electron wavefunction $\psi(x)$,

$$\frac{d^2\psi}{dx^2} + \frac{2m_e}{\hbar^2}\left[E - V(x)\right]\psi = 0 \qquad \textbf{(3.4.1)}$$

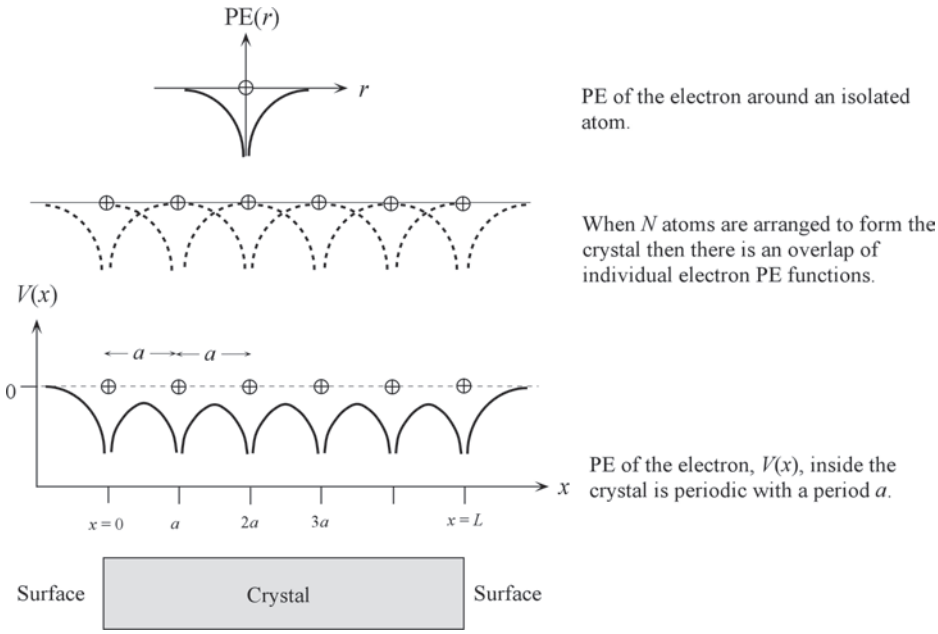


FIGURE 3.12 The electron potential energy (PE), $V(x)$, inside the crystal is periodic with the same periodicity as that of the crystal, a. Far away outside the crystal, by choice, $V = 0$ (the electron is free and PE $= 0$).

subject to the condition that the potential energy, $V(x)$, is periodic in a, that is,

$$V(x) = V(x + ma); \quad m = 1, 2, 3, \ldots \tag{3.4.2}$$

Periodic PE in the crystal

The solution of Eq. (3.4.1) will give the electron wavefunction in the crystal and hence the electron energy. Since $V(x)$ is periodic, we should expect, by intuition at least, the solution $\psi(x)$ to be periodic. It turns out that the solutions to Eq. (3.4.1), which are called **Bloch wavefunctions**, are of the form

$$\psi_k(x) = U_k(x)\exp(jkx) \tag{3.4.3}$$

Bloch wave in the crystal

where $U_k(x)$ is a periodic function which depends on $V(x)$ and has the same periodicity a as $V(x)$. The term $\exp(jkx)$, of course, represents a traveling wave whose wave vector is k. We should remember that we have to multiply this by $\exp(-jEt/\hbar)$, where E is the energy, to get the overall wavefunction $\Psi(x, t)$. Thus, the electron wavefunction in the crystal is a traveling wave which is modulated by $U_k(x)$. Further, both $\exp(jkx)$ and $\exp(-jkx)$ are possible and represent left and right traveling waves.

There are many such Bloch wavefunction solutions to the one-dimensional crystal, each identified with a particular k value, say k_n, which acts as a kind of quantum number. Each $\psi_k(x)$ solution corresponds to a particular k_n and represents **a state** with an energy E_k. The dependence of the energy E_k on the wave vector k is illustrated in what is called an ***E–k* diagram**. Figure 3.13 shows a typical *E–k* diagram for the hypothetical one-dimensional solid for k values in the range $-\pi/a$ to $+\pi/a$. Just as $\hbar k$ is the momentum of a free electron, $\hbar k$ for the Bloch electron is the momentum involved in its interaction with external fields, for example, those involved in the photon absorption processes. Indeed the rate of change of $\hbar k$ is the externally applied force F_{ext} on

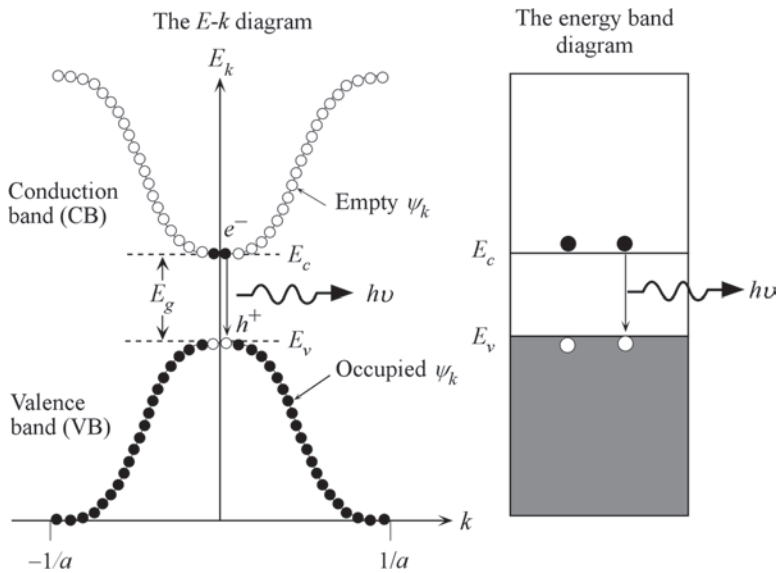


FIGURE 3.13 The *E–k* diagram of a direct bandgap semiconductor such as GaAs. The *E–k* curve consists of many discrete points with each point corresponding to a possible state, wavefunction $\psi_k(x)$, that is allowed to exist in the crystal. The points are so close that we normally draw the *E–k* relationship as a continuous curve. In the energy range E_v to E_c there are no points, that is, no $\psi_k(x)$ solutions.

the electron such as that due to an electric field E (*i.e.*, $F_{\text{ext}} = eE$). Thus, for the electron within the crystal, $d(\hbar k)/dt = F_{\text{ext}}$ and consequently we call $\hbar k$ the **crystal momentum** of the electron.[4]

Inasmuch as the momentum of the electron in the *x*-direction in the crystal is given by $\hbar k$, the *E–k* diagram is an energy vs. crystal momentum plot. The states $\psi_k(x)$ in the lower *E–k* curve constitute the wavefunctions for the valence electrons and thus correspond to the states in the valence band. Those in the upper *E–k* curve, on the other hand, correspond to the states in the conduction band since they have higher energies. All the valence electrons at 0 K therefore fill the states (particular k_n values) in the lower *E–k* diagram.

It should be emphasized that an *E–k* curve in the diagram consists of many discrete points, each point corresponding to a possible state, wavefunction $\psi_k(x)$, that is allowed to exist in the crystal. The points are so close that we draw the *E–k* relationship as a continuous curve. It is clear from the *E–k* diagram that there is a range of energies, from E_v to E_c, for which there are no solutions to the Schrödinger equation and hence there are no $\psi_k(x)$ with energies in E_v to E_c. Furthermore, we also note that the *E–k* behavior is not a simple parabolic relationship except near the bottom of the CB and the top of the VB.

Above absolute zero of temperature, due to thermal excitation, however, some of the electrons from the top of the valence band will be excited to the bottom of the conduction band. According to the *E–k* diagram in Figure 3.13, when an electron and hole recombine, the electron simply drops from the bottom of the CB to the top of the VB without any change in its *k* value so that this transition is quite acceptable in terms of momentum conservation. We should recall that the momentum of the emitted photon is negligible compared with the momentum of the electron. The *E–k* diagram in Figure 3.13 is therefore for a **direct bandgap semiconductor**. The minimum of the CB is directly above the maximum of the VB. As shown in Figure 3.13, the electron transition from the bottom of the CB to the top of the VB emits a photon.

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[4]The actual momentum of the electron, however, is not $\hbar k$ because $d(\hbar k)/dt \neq F_{\text{external}} + F_{\text{internal}}$. The true momentum p_e satisfies $dp_e/dt = F_{\text{external}} + F_{\text{internal}}$ (all forces on the electron). However, as we are interested in interactions with external forces such as an applied field, we treat $\hbar k$ as if it were the momentum of the electron in the crystal and use the name *crystal momentum.*

The simple *E–k* diagram sketched in Figure 3.13 is for the hypothetical one-dimensional crystal in which each atom simply bonds with two neighbors. In real crystals, we have a three-dimensional arrangement of atoms with $V(x, y, z)$ showing periodicity in more than one direction. The *E–k* curves are then not as simple as that in Figure 3.13 and often show unusual features. The *E–k* diagram for GaAs, which is shown in Figure 3.14 (a), as it turns out, has general features that are quite similar to that sketched in Figure 3.13. The lowest CB minimum is right above the top of the VB at the same *k*-value. GaAs is therefore a direct bandgap semiconductor in which electron–hole pairs can recombine directly and emit a photon. The majority of light-emitting devices use direct bandgap semiconductors to make use of direct recombination.

In the case of Si, the diamond crystal structure leads to an *E–k* diagram which has the essential features illustrated in Figure 3.14 (b). We notice that the minimum of the CB is *not* directly above the maximum of the VB, but it is displaced on the *k*-axis. Such crystals are called **indirect bandgap semiconductors**. An electron at the bottom of the CB cannot therefore recombine directly with a hole at the top of the VB, because for the electron to fall down to the top of the VB its momentum must change from k_{CB} to k_{VB}, which is not allowed by the law of conservation of momentum. Thus, direct electron–hole recombination does not normally take place in Si and Ge. The recombination process in these elemental semiconductors occurs via a **recombination center** at an energy level E_r within the bandgap as illustrated in Figure 3.14 (c). These recombination centers may be crystal defects or impurities. The electron is first captured by the defect at E_r. The change in the energy and momentum of the electron by this capture process is transferred to lattice vibrations, that is, to **phonons**. As much as an electromagnetic radiation is quantized in terms of photons, lattice vibrations in the crystal are quantized in terms of phonons. Lattice vibrations travel in the crystal just like a wave and these waves are called phonons. The captured electron at E_r can readily fall down into an empty state at the top of the VB and thereby recombine with a hole as in Figure 3.14 (c). Typically, the electron transition from E_r to E_v involves the emission of further lattice vibrations.

In some indirect bandgap semiconductors such as GaP, however, the recombination of the electron with a hole at certain recombination centers results in photon emission. The *E–k* diagram is similar to that shown in Figure 3.14 (c) except that the recombination centers at E_r are generated by the purposeful addition of nitrogen (N) impurities to GaP, written as GaP:N. The electron transition from E_r to E_v involves photon emission in the green.

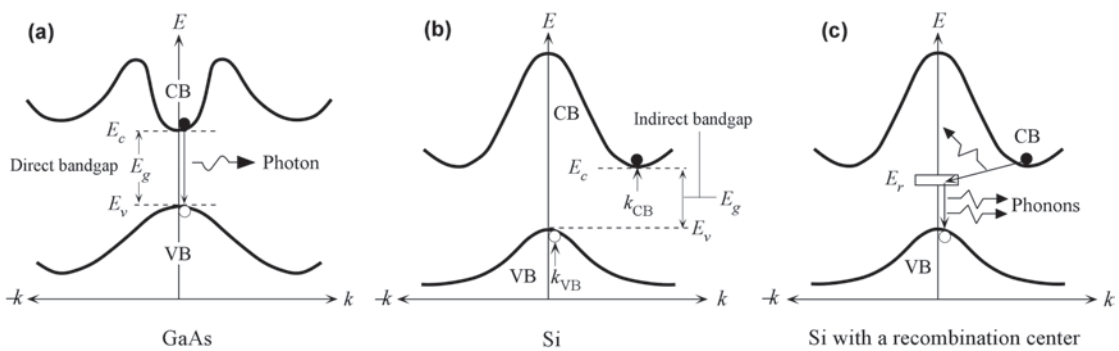


FIGURE 3.14 (a) In GaAs the minimum of the CB is directly above the maximum of the VB. GaAs is therefore a direct bandgap semiconductor. (b) In Si, the minimum of the CB is displaced from the maximum of the VB and Si is an indirect bandgap semiconductor. (c) Recombination of an electron and a hole in Si involves a recombination center.

3.5 *pn* JUNCTION PRINCIPLES

A. Open Circuit

Consider what happens when one side of a sample of Si is doped *n*-type and the other *p*-type as shown in Figure 3.15 (a). Assume that there is an *abrupt* discontinuity between the *p* and *n* regions which we call the **metallurgical junction**, *M*, as indicated in Figure 3.15 (a). There are fixed (immobile) ionized donors and free electrons (in the conduction band, CB) in the *n*-region and fixed ionized acceptors and holes (in the valence band, VB) in the *p*-region.

Due to the hole concentration gradient from the $p = p_{po}$ to the *n*-side where $p = p_{no}$, holes *diffuse* toward the right and enter the *n*-region and recombine with the electrons (majority carriers)

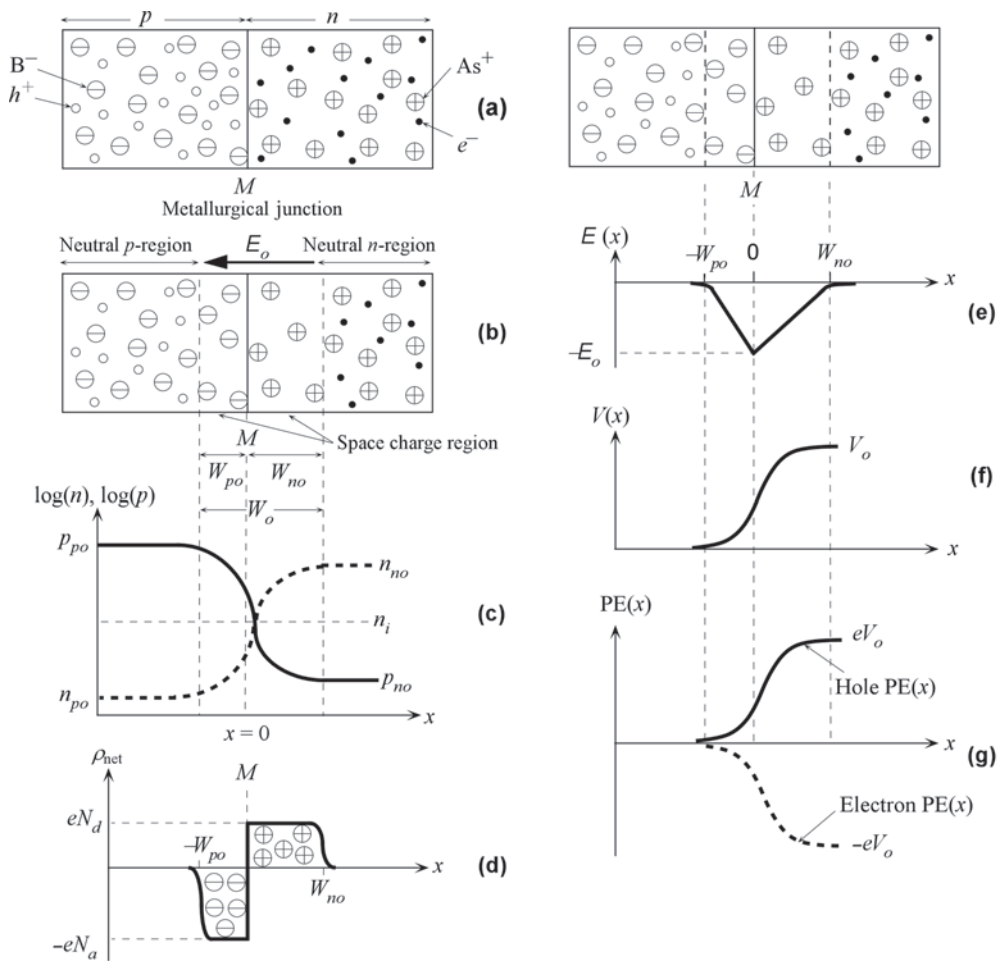


FIGURE 3.15 Properties of the *pn* junction. (a) The *p*- and *n*-sides of the *pn* junction before the contact. (b) The *pn* junction after contact, in equilibrium and in open circuit. (c) Carrier concentrations along the whole device through the *pn* junction. At all points, $n_{po}p_{po} = n_{no}p_{no} = n_i^2$. (d) Net space charge density ρ_{net} across the *pn* junction. (e) The electric field across the *pn* junction is found by integrating ρ_{net} in (d). (f) The potential $V(x)$ across the device. Contact potentials are not shown at the semiconductor–metal contacts. (g) Hole and electron potential energy (PE) across the *pn* junction. Potential energy is charge × potential = qV.

in this region. The *n*-side near the junction therefore becomes depleted of majority carriers and therefore has exposed positive donor ions (As^+) of concentration N_d. Similarly, the electron concentration gradient drives the electrons by diffusion toward the left. Electrons diffusing into the *p*-side recombine with the holes (majority carriers), which exposes negative acceptor ions (B^-) of concentration N_a in this region. The regions on both sides of the junction *M* consequently become *depleted* of free carriers in comparison with the bulk *p*- and *n*-regions far away from the junction. There is therefore a **space charge layer** (SCL) around *M*. Figure 3.15 (b) shows the SCL, also known as the **depletion region**, around *M*. Figure 3.15 (c) illustrates the hole and electron concentration profiles in which the vertical concentration scale is logarithmic. Note that we must, under equilibrium conditions (*e.g.*, no applied bias or photoexcitation), have $pn = n_i^2$ everywhere. The depletion region extends from about $x = -W_{po}$ to $x = W_{no}$. (The subscripts indicate the *p*- or *n*-side and open circuit.). The total width of the depletion region is $W_o = W_{po} + W_{no}$.

It is clear that there is an internal electric field E_o from positive ions to negative ions, that is, in the $-x$ direction, which tries to drift the holes back into the *p*-region and electrons back into the *n*-region. This field drives the holes in the opposite direction to their diffusion. As shown in Figure 3.15 (b), E_o imposes a drift force on holes in the $-x$ direction whereas the hole diffusion flux is in the $+x$ direction. A similar situation also applies for electrons with the electric field attempting to drift the electrons against diffusion from *n*- to the *p*-region. It is apparent that as more and more holes diffuse toward the right, and electrons toward the left, the internal field around *M* will increase until eventually an *equilibrium* is reached when the rate of holes diffusing toward the right is just balanced by the rates of holes drifting back to the left, driven by the field E_o. The electron diffusion and drift fluxes will also be balanced in equilibrium. For uniformly doped *p*- and *n*-regions, the net space charge density $\rho_{\text{net}}(x)$ across the semiconductor will be as shown in Figure 3.15 (d). The net space charge density ρ_{net} is negative and equal to $-eN_a$ in the SCL from $x = -W_{po}$ to $x = 0$ (*M* is at $x = 0$) and then positive and equal to $+eN_d$ from $x = 0$ to W_{no}. The total charge on the left-hand side must equal to that on the right-hand side for overall charge neutrality, so that

$$N_a W_{po} = N_d W_{no} \tag{3.5.1}$$

Depletion region widths

Figure 3.15 arbitrarily assumes that the acceptor concentration is greater than the donor concentration, $N_a > N_d$. From Eq. (3.5.1) this implies that $W_{no} > W_{po}$, that is, the depletion region penetrates the *n*-side (lightly doped side) more than the *p*-side (heavily doped side). Indeed, if $N_a \gg N_d$, then the depletion region is almost entirely on the *n*-side. We generally indicate heavily doped regions with the superscript plus sign as p^+. The electric field $E(x)$ and the net space charge density $\rho_{\text{net}}(x)$ at a point are related in electrostatics[5] by $dE/dx = \rho_{\text{net}}(x)/\varepsilon$, where $\varepsilon = \varepsilon_o \varepsilon_r$, is the permittivity of the medium, and ε_o and ε_r are the absolute permittivity and relative permittivity of the semiconductor material, respectively. We can thus integrate $\rho_{\text{net}}(x)$ across the device and thus determine the electric field. The variation of the electric field across the *pn* junction is shown in Figure 3.15 (e). The negative field means that it is in the $-x$ direction. Note that $E(x)$ reaches a maximum magnitude E_o at *M*.

The potential $V(x)$ at any point x can be found by integrating the electric field since by definition $E = -dV/dx$. Taking the potential on the *p*-side far away from *M* as zero (we have no applied voltage), which is an arbitrary reference level, $V(x)$ increases in the depletion region

[5]This is called *Gauss's law in point* form and comes from Gauss's law in electrostatics. The integration of the electric field *E* over a closed surface *S* is related to the total charge Q_{enclosed} enclosed, $\int E dS = Q_{\text{enclosed}}/\varepsilon$, where ε is the permittivity of the medium.

toward the *n*-side as indicated in Figure 3.15 (f). Notice that on the *n*-side the potential reaches V_o, which is called the **built-in potential**.

In an abrupt *pn* junction $\rho_{\text{net}}(x)$ can simply and approximately be described by step functions as displayed in Figure 3.15 (d). Using the step form of $\rho_{\text{net}}(x)$ in Figure 3.15 (d) and integrating it gives the electric field E_o and the built-in potential V_o,

Built-in field

$$E_o = -\frac{eN_dW_{no}}{\varepsilon} = -\frac{eN_aW_{po}}{\varepsilon} \tag{3.5.2}$$

and

Built-in potential

$$V_o = -\frac{1}{2}E_oW_o = -\frac{eN_aN_dW_o^2}{2\varepsilon(N_a + N_d)} \tag{3.5.3}$$

where $\varepsilon = \varepsilon_o\varepsilon_r$ is the permittivity of the semiconductor medium and $W_o = W_{no} + W_{po}$ is the total width of the depletion region under open circuit. If we know W_o, then W_{no} or W_{po} follows readily from Eq. (3.5.1). Equation (3.5.3) is a relationship between the built-in voltage V_o and the depletion region width W_o. If we know V_o we can calculate W_o.

The simplest way to relate V_o to the doping parameters is through Boltzmann statistics. For the system consisting of *p*- and *n*-type semiconductors together (forming one system), in equilibrium, the Boltzmann statistics[6] demands that the concentrations n_1 and n_2 of carriers at potential energies E_1 and E_2 be related by

Boltzmann statistics

$$n_2/n_1 = \exp\left[-(E_1 - E_2)/k_BT\right] \tag{3.5.4}$$

in which E is the potential energy—that is, qV, where q is charge and V voltage. Considering electrons, $q = -e$, we see from Figure 3.15 (g) that $E = 0$ on the *p*-side far away from M where $n = n_{po}$; and $E = -eV_o$ on the *n*-side away from M where $n = n_{no}$. Thus,

$$n_{po}/n_{no} = \exp(-eV_o/k_BT) \tag{3.5.5a}$$

Equation (3.5.5a) shows that V_o depends on n_{no} and n_{po} and hence on N_d and N_a. The corresponding equation for hole concentrations is

$$p_{no}/p_{po} = \exp(-eV_o/k_BT) \tag{3.5.5b}$$

Thus, rearranging Eqs. (3.5.5a) and (3.5.5b) we obtain

$$V_o = (k_BT/e)\ln(n_{no}/n_{po}) \quad \text{and} \quad V_o = (k_BT/e)\ln(p_{po}/p_{no})$$

We can now write p_{po} and p_{no} in terms of the dopant concentrations inasmuch as $p_{po} = N_a$, $p_{no} = n_i^2/n_{no} = n_i^2/N_d$, so that V_o becomes

Built-in potential

$$V_o = \frac{k_BT}{e}\ln\left(\frac{N_aN_d}{n_i^2}\right) \tag{3.5.6}$$

---

[6] We use Boltzmann statistics, *i.e.*, $n(E) \propto \exp(-E/k_BT)$, because the concentration of electrons in the conduction band whether on the *n*- or *p*-side is never so large that the Pauli exclusion principle becomes important. As long as the carrier concentration in the conduction band is much smaller than N_c, we can use Boltzmann statistics.

Clearly V_o has been conveniently related to the dopant and parent material properties via N_a, N_d, and n_i^2, which is given by $(N_c N_v)\exp(-E_g/k_B T)$. The built-in voltage ($V_o$) is the potential across a *pn* junction, going from *p*- to *n*-type semiconductor, in an open circuit. V_o is not the actual voltage across the diode. The voltage across the diode is made up of V_o and the contact potentials at the metal-to-semiconductor junctions at the electrodes. If we add V_o and the contact potentials at the electrode ends, we will find zero. Once we know the built-in potential V_o from Eq. (3.5.6), we can then calculate the width W_o of the depletion region from Eq. (3.5.3). The term $k_B T/e$ in Eq. (3.5.6) is called the **thermal voltage** V_{th}, and is about 26 mV at 300 K.

B. Forward Bias and the Shockley Diode Equation

Consider what happens when a battery with a voltage V is connected across a *pn* junction so that the positive terminal of the battery is attached to the *p*-side and the negative terminal to the *n*-side (forward bias). The negative polarity of the supply will reduce the potential barrier V_o by V, as shown in Figure 3.16 (a). The reason is that the bulk regions outside the SCL have high conductivities, due to the plenty of majority carriers in the bulk, in comparison with the depletion region in which there are mainly immobile ions. Thus, the applied voltage drops mostly across the depletion width W. The applied bias V now directly opposes V_o. The potential barrier against diffusion therefore becomes reduced to $(V_o - V)$ as illustrated in Figure 3.16 (b). This has drastic consequences because the probability that a hole in the *p*-side will surmount this potential barrier and diffuse to the *n*-side now becomes proportional to $\exp\left[-e(V_o - V)/k_B T\right]$. In other words, the applied voltage effectively reduces the built-in potential and hence the built-in field which acts against diffusion. Consequently, many holes can now diffuse across the depletion region and enter the *n*-side. This results in the **injection of excess minority carriers**, that is, holes into the *n*-region. Similarly, excess electrons can now diffuse toward the *p*-side and enter this region and thereby become injected minority carriers.

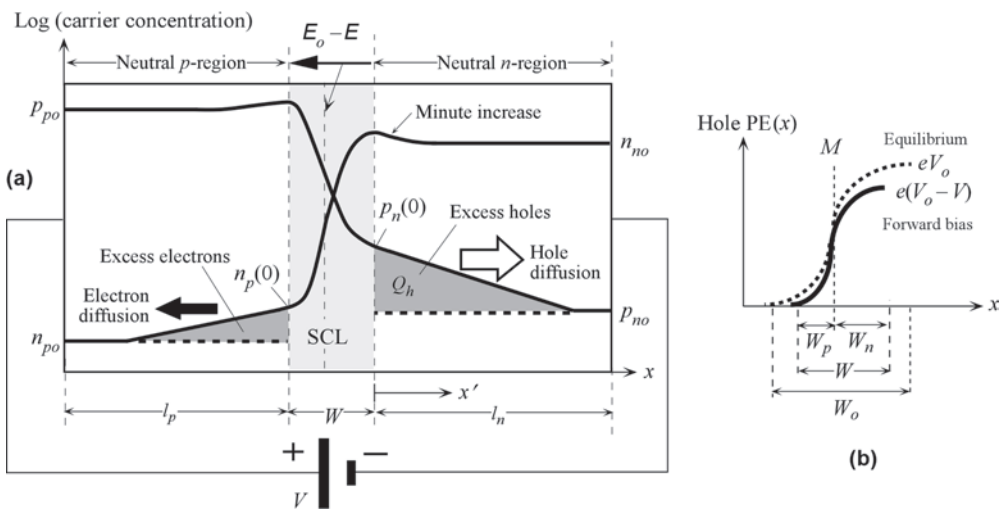


FIGURE 3.16 Forward-biased *pn* junction and the injection of minority carriers (a) Carrier concentration profiles across the device under forward bias. (b) The hole potential energy with and without an applied bias. W is the width of the SCL with forward bias.

The hole concentration $p_n(0)$ just outside the depletion region at $x' = 0$ (x' is measured from W_n) in the n-side is very large due to the injection of minority carriers as shown in Figure 3.16 (a). The hole concentration at the end of the neutral n-side at $x' = l_n$ at the negative terminal is always the equilibrium concentration p_{no}, because any additional holes at $x' = l_n$ would be neutralized immediately by electrons flowing from the negative terminal of the battery. It is obvious that there is a very large hole concentration gradient in the n-side as a result of minority carrier injection under forward bias, as shown in Figure 3.16 (a). The hole concentration gradient causes hole diffusion flux toward the negative terminal, and hence gives rise to an electric current. There is a similar electron diffusion flux and hence an electric current in the p-side. These two minority carrier diffusion fluxes constitute the forward bias pn junction current.

When holes are injected into the neutral n-side, they draw some electrons from the bulk of n-side (and hence from the battery) so that there is a small increase in the electron concentration. This small increase in the majority carriers is necessary to balance the hole charges and maintain neutrality in the n-side.

We now derive the pn junction equation. Consider the hole concentration profile in the n-side. The hole concentration, $p_n(0) = p_n(x' = 0)$, just outside the depletion region is due to the excess of holes diffusing through the SCL as a result of the reduction in the built-in potential barrier. This concentration, $p_n(0)$, is determined by the probability of surmounting the new potential energy barrier $e(V_o - V)$,

Boltzmann statistics and applied voltage

$$p_n(0) = p_{po} \exp\left[-\frac{e(V_o - V)}{k_B T}\right] \tag{3.5.7}$$

This follows directly from the Boltzmann equation, by virtue of the hole potential energy changing by $e(V_o - V)$ from $x = -W_p$ to $x = W_n$, as indicated in Figure 3.16 (b), and at the same time the hole concentration falling from p_{po} to $p_n(0)$. By dividing Eq. (3.5.7) by Eq. (3.5.5b), we get the effect of the applied voltage out directly, which shows how the voltage V determines the amount of *excess* holes diffusing and arriving at the n-region,

Law of the junction

$$p_n(0) = p_{no} \exp\left(\frac{eV}{k_B T}\right) \tag{3.5.8}$$

which is called the **law of the junction**. Equation (3.5.8) describes the effect of the applied voltage V on the injected minority carrier concentration just outside the depletion region, $p_n(0)$. Obviously, with no applied voltage, $V = 0$ and $p_n(0) = p_{no}$ which is exactly what we expect, Eq. (3.5.5b).

Injected holes diffuse in the n-region and eventually recombine with electrons in this region; there are many electrons in the n-side. Those electrons lost by recombination are readily replenished by the negative terminal of the battery connected to this side. The current due to holes diffusing in the n-region can be sustained because more holes can be supplied by the p-region, which itself can be replenished by the positive terminal of the battery.

Electrons are similarly injected from the n-side to the p-side. The electron concentration $n_p(0)$ just outside the depletion region at $x = -W_p$ is given by the equivalent of Eq. (3.5.8) for electrons, that is

Law of the junction

$$n_p(0) = n_{po} \exp\left(\frac{eV}{k_B T}\right) \tag{3.5.9}$$

In the *p*-region, the injected electrons diffuse toward the positive terminal looking to be collected. As they diffuse, they recombine with some of the many holes in this region. Those holes lost by recombination can be readily replenished by the positive terminal of the battery connected to this side. The current due to the diffusion of electrons in the *p*-side can be maintained by the supply of electrons from the *n*-side, which itself can be replenished by the negative terminal of the battery. It is apparent that an electric current can be maintained through a *pn* junction under forward bias, and that the current flow seems to be due to the *diffusion of minority carriers.* There is, however, some drift of majority carriers as well. The semiconductor *pn* junction is a minority carrier device in the sense that minority carriers play an important role.

When a hole (minority carrier) is injected into the *n*-side, it will diffuse in this region inasmuch as there is very little electric field in the neutral *n*-region to give rise to drift. Eventually the hole will recombine with an electron (majority carrier). The average time it takes for a hole (minority carrier) to recombine with an electron in the *n*-side is called the **minority carrier recombination time** or **lifetime** τ_h. The reciprocal $1/\tau_h$ is the mean probability per unit time that a hole will recombine and disappear. The ability of holes to diffuse in a semiconductor is represented by their **diffusion coefficient** D_h, which is related the their drift mobility μ_h through the Einstein relation.[7] An average distance a hole diffuses in the *n*-side before it recombines with an electron is called the **diffusion length** L_h and is given by $L_h = (D_h\tau_h)^{1/2}$. We can interpret the reciprocal $1/L_h$ in a similar way to $1/\tau_h$ as the probability per unit distance that a hole recombines with an electron in the *n*-side as it diffuses along *x*.

If the length of the *n*-region l_n is much longer than the minority carrier diffusion length L_h, most (nearly all) injected holes will eventually recombine before they reach the negative electrode. The hole concentration profile $p_n(x')$ vs. x' therefore decays from $p_n(0)$ toward the thermal equilibrium value, p_{no}, as illustrated in Figure 3.16 (a). Consider $\Delta p_n(x') = p_n(x') - p_{no}$, which is the **excess carrier concentration**, then the change $\delta\Delta p_n$ in excess concentration over a distance δx due to recombination would depend on Δp_n itself (how many excess holes we have) and also on the probability of recombination $(\delta x')(1/L_h)$ so that $\delta\Delta p_n = -\Delta p_n(\delta x'/L_h)$, which implies an exponential decay,

Long diode minority carrier profile

$$\Delta p_n(x') = \Delta p_n(0)\exp(-x'/L_h) \tag{3.5.10}$$

Equation (3.5.10) can be derived by rigorous means using the Continuity Equation[8] rather than the above intuitive arguments. The *pn* junction in which $l_n \gg L_h$ is called a **long diode**.

The hole diffusion current density $J_{D,\text{hole}}$ is the *hole diffusion flux* multiplied by the hole charge,[9]

$$J_{D,\text{hole}} = -eD_h\frac{dp_n(x')}{dx'} = -eD_h\frac{d\Delta p_n(x')}{dx'}$$

[7] In nondegenerate semiconductors, the diffusion coefficient D and the drift mobility μ for a given type of carrier are related by $D/\mu = kT/e$ so that we can easily calculate D from μ. The proof is not difficult; see, for example, S. O. Kasap, *Principles of Electronic Materials and Devices*, 3rd Edition (McGraw-Hill, 2006), Ch. 5.

[8] The intuitive derivation here for Eq. (3.5.10) was actually based on knowing the outcome from the continuity equation and using the fact that we have a very long *n*-side. In the latter case *only*, the oversimplified intuitive arguments above do indeed lead to Eq. (3.5.10); but not in a general treatment. One must also consider the change in the diffusion flux (the current). (See, for example, S. O. Kasap, *Principles of Electronic Materials and Devices*, 3rd Edition, McGraw-Hill, 2006), Ch. 6.

[9] The hole diffusion flux is $-D_h(dp/dx) = -D_h(d\Delta p/dx)$ and the diffusing charge is $+e$.

i.e.,

$$J_{D,\text{hole}} = \left(\frac{eD_h}{L_h}\right)\Delta p_n(0)\exp\left(-\frac{x'}{L_h}\right) \tag{3.5.11}$$

The above equation shows that hole diffusion current depends on location x' and decreases due to recombination. The total current at any location, however, is the sum of hole and electron contributions. The total current is independent of x' as indicated in Figure 3.17. The decrease in the minority carrier diffusion current with x' is made up by the increase in the current due to the drift of the majority carriers as schematically shown in Figure 3.17. The field in the neutral region is not totally zero but a small value, just sufficient to drift the huge number of majority carriers there and maintain a constant current.

We can now use the law of the junction to substitute for $\Delta p_n(0)$ in Eq. (3.5.11) in terms of the applied voltage V in Eq. (3.5.8). Further, we can eliminate p_{no} by $p_{no} = n_i^2/n_{no} = n_i^2/N_d$. Thus, at $x' = 0$, just outside the depletion region, from Eq. (3.5.11) the hole diffusion current density is

Hole diffusion current

$$J_{D,\text{hole}} = \left(\frac{eD_h n_i^2}{L_h N_d}\right)\left[\exp\left(\frac{eV}{k_B T}\right) - 1\right] \tag{3.5.12}$$

There is a similar expression for the electron diffusion current density $J_{D,\text{elec}}$ in the p-region. We will assume that the electron and hole currents do not change across the depletion region because, in general, the width of this region is quite narrow (and, for the time being, we neglect the recombination in the SCL). The electron current at $x = -W_p$ is the same as that at $x = W_n$. The total current density is then simply given by $J_{D,\text{hole}} + J_{D,\text{elec}}$, that is,

Shockley long diode equation

$$J = \left(\frac{eD_h}{L_h N_d} + \frac{eD_e}{L_e N_a}\right)n_i^2\left[\exp\left(\frac{eV}{k_B T}\right) - 1\right] \tag{3.5.13a}$$

or

Shockley long diode equation

$$J = J_{so}\left[\exp\left(\frac{eV}{k_B T}\right) - 1\right] \tag{3.5.13b}$$

This is the familiar diode equation with $J_{so} = \left[(eD_h/L_h N_d) + (eD_e/L_e N_a)\right]n_i^2$. It is frequently called the **Shockley equation**. It represents the *diffusion of minority carriers* in the neutral regions. The constant J_{so} depends not only on the doping, N_d, N_a, but also on the material

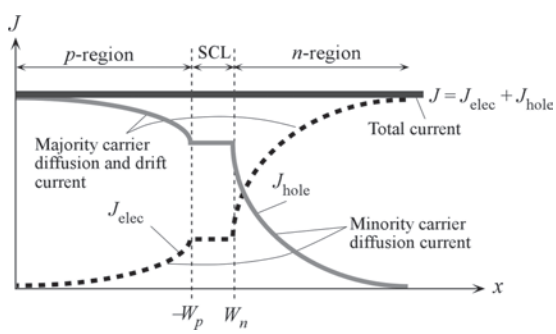


FIGURE 3.17 The total current anywhere in the device is constant. Just outside the depletion region it is primarily due to the diffusion of minority carriers and near the contacts it is primarily due to the drift of majority carriers.

via n_i, D_h, D_e, L_h, L_e. It is known as the **reverse saturation current density**, because if we apply a reverse bias $V = -V_r$ greater than the thermal voltage k_BT/e (25 mV), Eq. (3.5.13b) becomes $J = -J_{so}$.

We note that since the applied voltage drops across the depletion region, the built-in field is reduced by the applied field imposed by the bias, that is, the field at the junction becomes $E_o - E$, where E is the applied field at the junction. Because the field is reduced, we need less charges on either side of the junction. This can only be accommodated if the depletion regions shrink in width to W_p and W_n on the p- and n-sides, and the new total width $W = W_p + W_n$ is less than W_o. Equation (3.5.1) remains valid in the form $N_aW_p = N_dW_n$.

The *pn* junction in which the neutral regions l_n and l_p are shorter than the diffusion lengths L_h and L_n, respectively, is called a **short diode**. Its treatment is much easier than the long diode. There is essentially no time for the injected hole (minority carrier) in the n-side to recombine because it quickly diffuses and reaches the collection (negative) electrode, which is very close in distance. If we are not losing holes by recombination, then there is no reason for the hole diffusion current to decay along x' in this region as in the long diode where recombination is present. The hole diffusion current would remain constant if the hole concentration gradient is constant, that is, the excess hole concentration profile decreases linearly as shown in Figure 3.18. Thus, the hole diffusion current density $J_{D,\text{hole}}$ on the n-side is

$$J_{D,\text{hole}} = -eD_h\frac{d\Delta p_n(x')}{dx'} = eD_h\frac{p_n(0) - p_{no}}{l_n} = eD_h\frac{p_{no}e^{eV/k_BT} - p_{no}}{l_n}$$

so that

$$J_{D,\text{hole}} = \frac{eD_hn_i^2}{l_nN_d}\left[\exp\left(\frac{eV}{k_BT}\right) - 1\right]$$

A similar argument applies for $J_{\text{D,elec}}$ on the p-side so that the total current density is then

$$J = \left(\frac{eD_h}{l_nN_d} + \frac{eD_e}{l_pN_a}\right)n_i^2\left[\exp\left(\frac{eV}{k_BT}\right) - 1\right] \qquad \textbf{(3.5.14)}$$

Shockley short diode equation

Equation (3.5.14) is identical to Eq. (3.5.13a) with diffusion lengths L_h and L_e replaced by the widths of the neutral regions l_n and l_p, respectively. The short diode is a good example of a

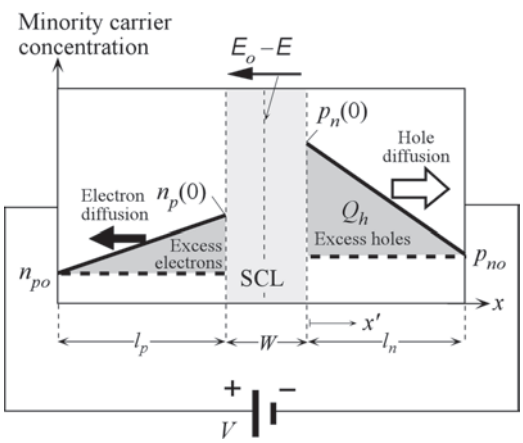


FIGURE 3.18 Minority carrier injection and diffusion in a short diode. Minority recombination in the neutral regions is negligible. $p_n(0)$ is the minority carrier concentration just outside the depletion region, and is controlled by the law of the junction, $p_n(0) = p_{no}\exp(eV/k_BT)$.

device in which the current is almost entirely due to the *diffusion of minority carriers*, and highlights the significant role played by diffusion.

C. Minority Carrier Charge Stored in Forward Bias

So far we have considered a *pn* junction with heavier doping on the *p*-side. The injected minority carriers, holes in this case, represent an *injected excess minority carrier charge*, Q_h, in the neutral region as shown in Figures 3.16 (a) and 3.18. There is, of course, also excess majority carrier charge so the region is neutral; but the excess majority carrier concentration (about the same as $p_n(0)$) is very small compared with the actual majority carrier concentration ($n_{no} = N_d$). For the long diode, the excess holes in the neutral *n*-side diffuse and recombine after τ_h seconds—the minority carrier lifetime. The charge Q_h should therefore disappear after a time τ_h. It must therefore be replenished at a rate Q_h/τ_h if Q_h is to be maintained at steady state DC operation. The external current supplies holes at exactly this rate to maintain the steady state DC conduction. Thus, $I = Q_h/\tau_h$.

In the case of a short diode, a hole that is injected into the *n*-side must diffuse across the width l_n of this neutral region to reach the negative electrode. The time required for holes to diffuse across the width l_n of the neutral *n*-region is called the **transit time** τ_t, or the **diffusion time**. From random walk examples in various materials textbooks,[10] τ_t is given by $\tau_t = l_n^2/2D_h$. To maintain DC operating conditions, the current must replenish holes exactly at a rate Q_h/τ_t to make up for those holes reaching the end of the *n*-side, that is, $I = Q_h/\tau_t$. We can now generalize the above discussion as follows:

DC currents

$$I = Q/\tau \text{ for a } \textbf{long diode} \quad \text{and} \quad I = Q/\tau_t \text{ for a } \textbf{short diode} \tag{3.5.15}$$

where τ is the **effective minority carrier lifetime**, $1/\tau = 1/\tau_h + 1/\tau_e$, and τ_t is the **effective transit time**, that is, $1/\tau_t = 1/\tau_{th} + 1/\tau_{te}$; subscripts refer to holes and electrons in the neutral *n*- and *p*-sides, respectively. Equation (3.5.15) applies under DC conditions only.

D. Recombination Current and the Total Current

So far we have assumed that, under a forward bias, the minority carriers diffusing and recombining in the neutral regions are supplied by the external current. However, some of the minority carriers will recombine in the depletion region. The external current must therefore also supply the carriers lost in the recombination process in the SCL, as illustrated in Figure 3.19 (a). Consider for simplicity a symmetrical *pn* junction as in Figure 3.19 (b) under forward bias. At the junction center *C*, the hole and electron concentrations are p_M and n_M and are equal. We can find the SCL recombination current by considering electrons recombining in the *p*-side in W_p and holes recombining in the *n*-side in W_n as shown by the shaded areas *ABC* and *BCD*, respectively, in Figure 3.19 (b). Suppose that the **mean hole recombination time** in W_n is τ_h and **mean electron recombination time** in W_p is τ_e. The rate at which the electrons in *ABC* are recombining is the area *ABC* (nearly all injected electrons) divided by τ_e. The electrons are replenished by the diode current. Similarly, the rate at which holes in *BCD* are recombining is the area *BCD* divided by τ_h. Thus, the recombination current density is

$$J_{\text{recom}} = \frac{eABC}{\tau_e} + \frac{eBCD}{\tau_h}$$

[10]The expression for the mean square distance (l^2) diffused in a time interval τ, $l^2 = 2D\tau$, is derived in S. O. Kasap, *Principles of Electronic Materials and Devices*, 3rd Edition (McGraw-Hill, 2006), Ch. 1.

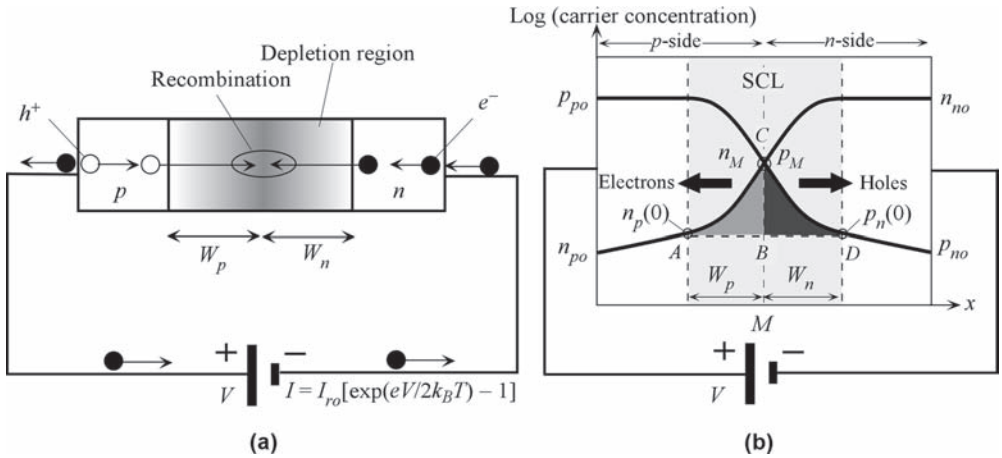


FIGURE 3.19 Forward-biased *pn* junction, the injection of carriers, and their recombination in the SCL. (a) Recombination of electrons and holes in the depletion region involves the current supplying the carriers. (b) A symmetrical *pn* junction for calculating the recombination current.

We can evaluate the areas *ABC* and *BCD* by taking them as triangles, $ABC \approx (1/2)W_p n_M$, *etc.*, so that

$$J_{\text{recom}} = \frac{e\frac{1}{2}W_p n_M}{\tau_e} + \frac{e\frac{1}{2}W_n p_M}{\tau_h} \tag{3.5.16}$$

Under steady state and equilibrium conditions, assuming a nondegenerate semiconductor, we can use Boltzmann statistics in Eq. (3.5.4) to relate these concentrations to the potential energy. At *A*, the hole potential energy is zero ($V = 0$) and at *M* it is $e(V_o - V)/2$ so that

$$\frac{p_M}{p_{po}} = \exp\left[-\frac{e(V_o - V)}{2k_BT}\right]$$

Since V_o depends on dopant concentrations and n_i as in Eq. (3.5.6) and further $p_{po} = N_a$, we can simplify the above to

$$p_M = n_i \exp\left(\frac{eV}{2k_BT}\right)$$

and there is a similar expression for n_M. We can substitute for p_M and n_M in Eq. (3.5.16) to find the recombination current for $V > k_BT/e$, that is,

$$J_{\text{recom}} = \frac{en_i}{2}\left(\frac{W_p}{\tau_e} + \frac{W_n}{\tau_h}\right)\exp\left(\frac{eV}{2k_BT}\right) \tag{3.5.17a}$$

Recombination current

From a better mathematical treatment, the expression for the recombination current can be shown to be[11]

$$J_{\text{recom}} = J_{ro}\left[\exp(eV/2k_BT) - 1\right] \tag{3.5.17b}$$

SCL recombination current

where J_{ro} is the pre-exponential constant in Eq. (3.5.17a).

[11] This is generally proved in advanced texts.

Equation (3.5.17) is the current that supplies the carriers that recombine in the depletion region. It does not stipulate what the recombination process is, but only the carriers recombine in this region. The recombination could be direct (as in high quality GaAs *pn* junctions) or indirect through defects or impurities in the SCL.

In many III–V *compound* semiconductors, the carriers also recombine at crystal surfaces and interfaces where, usually, there are defects that encourage recombination; or act as recombination centers. The external current also has to supply carriers that are lost through surface recombination. The exact *I–V* dependence for surface recombination is difficult to derive but has a voltage dependence quite similar to recombination in the SCL in Eq. (3.5.17b). Thus, surface recombination results in a current that can be described by

Surface recombination current

$$I_{\text{surf}} = I_{sro}\left[\exp\left(eV/2k_BT\right) - 1\right] \tag{3.5.18}$$

where I_{sro} is the pre-exponential constant that depends on the surface recombination process, and the surface area involved in the recombination.

The total current into the diode will supply carriers for minority carrier diffusion in the neutral regions and recombination in the space charge layer and at the surfaces and interfaces so that it will *be the sum* of Eqs. (3.5.13b), (3.5.17b), and (3.5.18). Generally the diode current is written as

General diode equation

$$I = I_o\left[\exp\left(\frac{eV}{\eta k_BT}\right) - 1\right] \tag{3.5.19}$$

where I_o is a constant and η, called the **diode ideality factor**, is 1 for diffusion-controlled and 2 for recombination-controlled forward current characteristics.

Figure 3.20 shows typical semilogarithmic plots of the forward *I–V* characteristics of three types of *pn* junctions based on Ge, Si, and GaAs. The Ge *pn* junction follows the Shockley model with $\eta = 1$ and the current is due to diffusion in the neutral regions. Both Si and GaAs diodes initially follow the recombination model and then the Shockley model. Many Si and GaAs diodes have an η that is between 1 and 2 for the *I–V* ranges of interest.

Notice that, at a given current, say 1 mA, the voltage across a Ge diode is about 0.2 V, it is 0.7 V across the Si diode, and almost 1.1 V across the GaAs diode. The increase in the turn-on

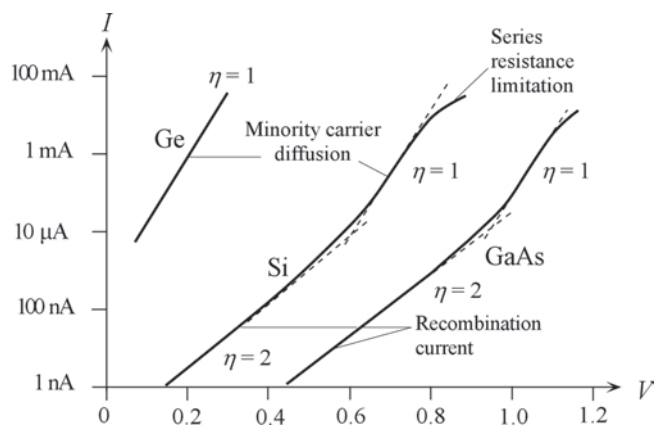


FIGURE 3.20 Schematic representation of the forward *I–V* characteristics of three typical *pn* junctions based on Ge, Si, and GaAs.

Upon excitation by an incident radiation of suitable energy $h\nu_{ex}$, the activator becomes excited to E_2'' in the E_2-manifold; a vertical transition. From this energy level, it decays, or *relaxes*, down relatively quickly (on a time scale of the order of picoseconds) to an energy level E_2' by emitting phonons or lattice vibrations. This type of decay is called **radiationless** or **non-radiative decay**. The Ce^{3+} ion becomes slightly displaced as shown in Figure 3.48 (a). From E_2', the activator decays down to E_1' in the E_1-manifold by emitting a photon by spontaneous emission, which is the emitted luminescent radiation. This is a vertical transition (Franck–Condon principle). The emitted photon energy is $h\nu_{em}$, which is less than the excitation photon energy $h\nu_{ex}$. The return from E_1' to the ground E_1 state involves phonon emissions. In some activators, the higher levels may form multilevel narrow energy "bands." In this example, the activator absorbed the incident radiation, and was directly excited, which is known as **activator excitation**. The Ce^{3+} ions in $Y_2Al_5O_{12}:Ce^{3+}$ can be excited directly by blue light, and would then emit in the yellow region as discussed below. It is apparent from Figure 3.48 (a) that the emitted radiation ($h\nu_{em}$) has a *longer* wavelength than the exciting radiation ($h\nu_{ex}$), that is, $h\nu_{em} < h\nu_{ex}$. The downshift in the light frequency from absorbed to emitted radiation is called the **Stokes shift**. It should be emphasized that the energy levels of the activator also depend on the host, because the internal electric fields within the host crystal act on the activator, and shift these levels up and down. The emission characteristics depend firstly on the activator, and secondly on the host.

Commercially available and popular white LEDs can provide bright light at a fraction of input electric power compared with conventional tungsten-based incandescent lights. In addition, the white LED has an extremely long lifetime. Most common white LEDs use a chip emitting at a short wavelength (blue or violet) and a *phosphor*, which absorbs some of the light from the blue diode and undergoes secondary luminescence emissions at a longer wavelength (yellow) as illustrated in Figure 3.48 (b). The mixture of blue light from the chip and yellow light from the phosphor results in an overall spectrum, shown in Figure 3.48 (c), that appears white to the eye. The quality and spectral characteristics of the combined emission in Figure 3.48 (c) vary with different white LED designs, for example, chip and phosphor combination.

Typically, a phosphor is composed of an inorganic host substance that contains optically active dopants; the dopants are the atoms that emit the lower energy photons, that is, the activators. Yttrium aluminum garnet (YAG) is a common host material. For white LED applications, it is usually doped with one of the rare-earth elements or a rare-earth compound. Cerium (Ce) is a common dopant element in YAG phosphors designed for white light–emitting diodes in



LUXEON Rebel ES white-emitting LED. *(Courtesy of Philips Lumileds.)*



Snap LED emitting in the amber for automotive signaling applications. The lamp includes the driver under the LED. *(Courtesy of Philips Lumileds.)*



The Audi A8 uses LEDs for nearly all its lighting, including headlights. *(Used with permission of Audi of America, Inc.)*

so-called **blue chip and yellow phosphor white LEDs**. White LEDs are now challenging the existing incandescent sources for general lighting.

In applications requiring a full spectrum of colors from a single point source, red, green, and blue (RGB)–emitting chips in a single package are used, which allows the generation of white light or any of 256 colors by utilizing circuits that drive the three diodes independently.

Additional Topics

3.18 LED ELECTRONICS

The electronics involved in driving LEDs depends on the application, which generally falls into three categories: (a) DC operation, (b) modulation of the LED output light intensity about some DC value in analog applications, and (c) pulsed operation in digital applications, including optical communications. In DC operation, the LED is driven by providing the necessary current for the required optical output. The *I–V* characteristics is such that large changes in the current correspond to small voltage changes across the LED as in Figure 3.43 (a), and for a given LED, V_F does not vary very much. The simplest circuit is shown in Figure 3.49 (a), where a DC voltage $V(>V_F)$ provides the current, which is determined by $I_F = (V - V_F)/R$. The problem is that keeping I_F constant depends on keeping V constant. Most LEDs have low reverse breakdown voltages and usually another diode, such as D, is used in parallel to shunt the LED when a reverse voltage is accidently applied to the LED.

A better circuit than that in Figure 3.49 (a) should provide the necessary drive current I_F to the LED, with some convenient way of controlling I_F. Bipolar junction transistors (BJTs) are commonly used in many applications as convenient current sources. Figure 3.49 (b) is shown as an example in which the LED is in the collector circuit of the BJT, which is driven by the output of an op amp A. The resistor R_F in the emitter of the BJT provides feedback, $I_F R_F$, to the negative input of A, which stabilizes the circuit. It is straightforward to show that $I_F \approx V/R_F$, that is, the circuit provides a linear control of I_F by the input voltage V. There are many ICs available that will perform the function of providing the necessary constant current under a wide range of input

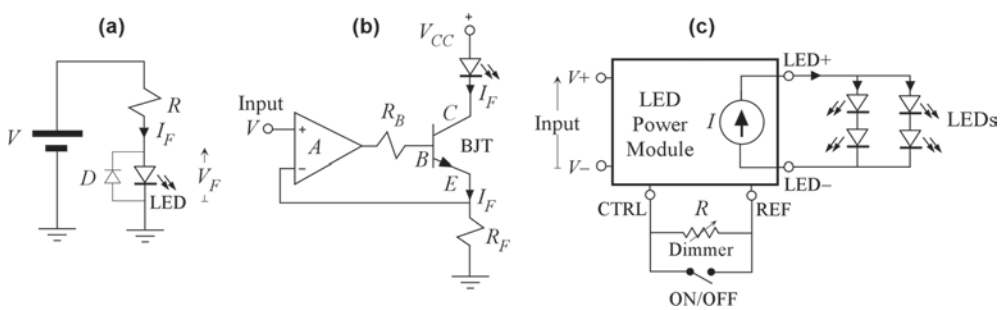


FIGURE 3.49 (a) The simplest circuit to drive an LED involves connecting it to a voltage supply (V) through a resistor R. (b). Bipolar junction transistors are well suited for supplying a constant current. Using an IC and negative feedback, the current is linearly controlled by V. (c) There are various commercial LED driver modules that can be easily configured to drive a number of LEDs in parallel and/or series. The example has a module driving four LEDs, a dimmer (R), and an on/off switch.

voltages, and temperature variations. Without a proper control of the current, the temperature variations can cause substantial changes in the current and hence the light output. Figure 3.49 (c) uses one of several LED driver modules available in the market, which can drive more than one LED in various parallel and series combinations, provided the maximum current rating is satisfied. The current can be externally and easily controlled, for example, by an external variable resistor, or a microprocessor.

The LED drive circuit shown in Figure 3.49 (b) can also be used in modulating the output of the LED as well since the current is proportional to the input voltage as in Figure 3.50 (a). One can, of course, design faster drive circuits, but the bottleneck will eventually be in the speed of response of the LED. An AC signal of frequency f on top of a DC voltage is used to drive the LED in Figure 3.50 (a). The frequency dependence of the output light power $P_o(f)$ with respect to its DC value $P_o(0)$ is shown in Figure 3.50 (b). There is a clear roll-off at high frequencies. The **cutoff frequency** f_c is defined as the frequency at which the normalized power drops by a factor of $2^{1/2}$, that is, at $f = f_c$, $P_o(f_c) = 0.707P_o(0)$. Any modulation of the LED current would result in modulating the minority carrier injection profile as shown in Figure 3.24 (a). The small signal equivalent circuit shown in Figure 3.24 (c) is usually sufficient to understand the frequency response of the LED but we have to be careful inasmuch as r_d and C_{diff} also depend on the frequency at high frequencies. First, consider very low frequencies, that is, almost DC. The capacitances are negligible. The diode action embedded in $I = I_o \exp(eV/k_BT)$ and hence the optical power output P_o, depend on the current in r_d. At high frequencies, the capacitances shunt away this current from r_d and decrease the optical output power. The characteristic time constant τ should be r_dC_{diff} (C_{diff} is greater than the depletion layer capacitance C_{dep}), that is, the net minority carrier recombination time. A more rigorous treatise arrives at the same conclusion and gives the output optical power $P_o(\omega)$ at frequency ω as

LED optical power output

$$P_o(\omega)/P_o(0) = 1/\left[1 + (\omega\tau)^2\right]^{1/2} \tag{3.18.1}$$

At the critical frequency $f_c = 1/(2\pi\tau)$ in Eq. (3.18.1), the relative output power is 0.707 as shown in Figure 3.50 (b). The **optical bandwidth** f_{op} is defined as the frequency at which $P_o(f_{\text{op}})/P_o(0) = 1/2$, and occurs when $f_{\text{op}} = \sqrt{3}f_c$, that is,

Optical bandwidth of an LED

$$f_{\text{op}} = \sqrt{3}/(2\pi\tau) \tag{3.18.2}$$

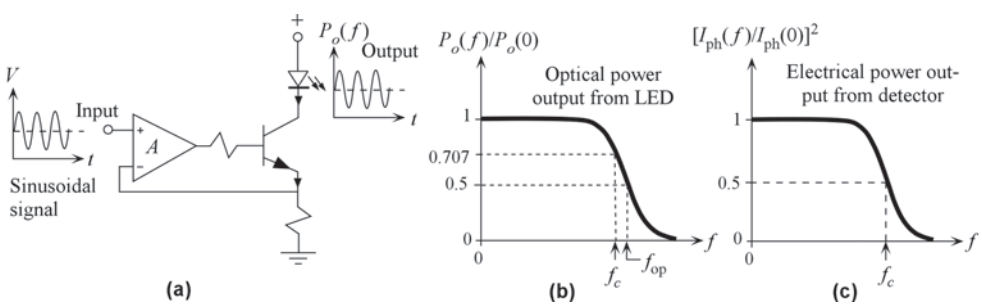


FIGURE 3.50 (a) Sinusoidal modulation of an LED. (b) The frequency response where f_c is the cutoff frequency at which $P_o(f)/P_o(0)$ is 0.707. (c) The electrical power output from the detector as a function of frequency. At f_c, $\left[I_{\text{ph}}(f)/I_{\text{ph}}(0)\right]^2$ is 0.5. However, it is 0.707 at a lower frequency than f_c.

where τ is the effective or net recombination time that is, $1/\tau = 1/\tau_r + \tau_{nr}$, where τ_r and τ_{nr} are the radiative and nonradiative recombination lifetimes. Recombination centers in the semiconductor reduce τ_{nr} and hence τ, which would widen the bandwidth. However, the IQE is also reduced, so that the optical output power is diminished as well, which is normally undesirable. If we feed the output from the LED to a photodetector, then the detector photocurrent I_{ph} will be proportional to P_o and the electrical power delivered by the detector will be proportional to I_{ph}^2 or P_o^2. At the frequency f_c, the detector's electrical output power will be proportional to $[0.707P_o(0)]^2$ or $0.5P_o(0)^2$. Thus, f_c represents a cutoff frequency (a bandwidth) at which the electrical power available from the detector has fallen to half its unmodulated value ($f = 0$ value), as illustrated in Figure 3.50 (c); f_c is called the **electrical bandwidth**.

The turning on and off of an LED in digital applications involves pumping the minority carriers into the active region and then removing them when the LED is turned off. In principle, the minority carrier lifetime τ should also play a key role in this pulsed operation. Figure 3.51 shows a simplified diagram for driving the LED digitally, for example, from a logic gate. The light output pulse should have an exponential rise and fall portions that delay the full turn-on and turn-off of the LED. The characteristic time constant for the rise and fall portions of the pulse should be the net recombination time τ. In practice, exponential rise and fall only serve as approximations to the switching behavior of the output light pulse. The net recombination time τ itself can depend on the current, and hence we should not expect a perfect exponential rise and fall.

Most LED data sheets quote the rise and fall times τ_R and τ_F for LEDs that have been pulsed. τ_R is the rise time from 10% to 90% of the final pulse height as shown in Figure 3.51. Similarly the fall time τ_F is the time it takes for the pulse to drop from its 90% to 10% value before the turn-off was triggered. If we approximate the rise and fall portions as exponentials as mentioned above, then, $\tau_R = \tau_F = 2.2\tau$. It is possible to shape the output pulse by intruding a pulse shaping capacitor C across the resistor R as shown in Figure 3.51. Upon triggering, the capacitor can provide the initial large current to pump (inject) the minority carriers in and hence turn on the device more quickly. Similarly, the capacitor would allow a large turn-off current to take out the minority carriers. Thus τ_R and τ_F can be shortened. The RC time constant must match the net recombination time in the LED for best reduction in the rise and fall times.

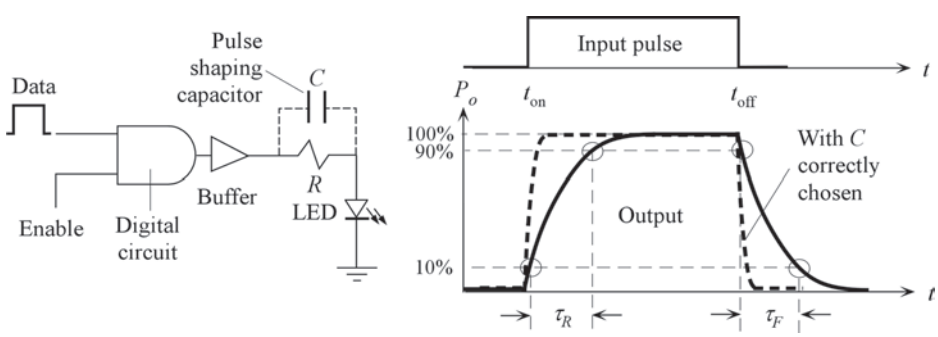


FIGURE 3.51 An LED in a digital circuit is turned on and off by a logic gate, assumed to have a buffered output as shown, to avoid being loaded. A BJT can be used after the logic gate to drive the LED as well (not shown). Definitions of rise and fall times are shown in the light output pulse.

Questions and Problems

3.1 Metals and work function The metal sodium (Na) has an atomic concentration of 2.54×10^{22} cm^{-3}. Each Na atom in isolation has one outer valence electron in an unfilled 3*s*-subshell. Once the solid is formed, the outer valence electrons from all Na atoms are shared in the crystal by all Na^+ ions, that is, these valence electrons find themselves in an unfilled energy band as in Figure 3.1 (c). Since each Na atom donates one electron to the energy band, the electron concentration is the same as the atomic concentration. Calculate the Fermi energy at 0 K. What is the speed of the electrons at E_F? What should be the speed at 300 K if the electrons could be treated classically in terms of the kinetic molecular theory as if they were free, similar to the atoms in a gas?

3.2 Photocathode and work function The photocathode of a photomultiplier tube has a multi-alkaline (Sb-Na-K-Cs) metal with a work function (Φ) of 1.55 eV. What is the longest wavelength that will cause photoemission? What is the kinetic energy of a photoemitted electron if the incident light wavelength is 450 nm (blue)? The quantum efficiency (QE) of a photocathode is defined by

$$\text{Quantum efficiency} = \frac{\text{Number of photoemitted electrons}}{\text{Number of incident photons}}$$

The QE is 100% if each incident photon ejects one electron. Suppose that blue light of wavelength 450 nm with an intensity of 1 μW cm^{-2} is incident on this photocathode with an area of 50 mm^2. If the emitted electrons are collected by applying a positive bias voltage to an anode, and the photocathode has a QE of 25%, what will be the photocurrent? (Normally the photoemitted electron is accelerated by a suitable applied field and impacts another electrode, a dynode, where it causes secondary electron emission, and so on, until the current is multiplied by orders of magnitude.)

3.3 Refractive index and bandgap Diamond, silicon, and germanium all have the same diamond unit cell. All three are covalently bonded solids. Their refractive indices (n) and energy bandgaps (E_g) are shown in Table 3.2. (a) Plot n vs. E_g and (b) Plot also n^4 vs. $1/E_g$. What is your conclusion? According to **Moss's rule**, roughly, $n^4 E_g \approx K$, a constant. What is the value of K?

TABLE 3.2 The refractive index n and the bandgap E_g of diamond, Ge, and Si, all of which have the same crystal structure

Material →	Diamond	Silicon	Germanium
Bandgap, E_g (eV)	5	1.1	0.66
n	2.4	3.46	4.0

3.4 Electrons in the CB of a nondegenerate semiconductor

(a) Consider the energy distribution of electrons $n_E(E)$ in the conduction band. Assuming that the density of state $g_{CB}(E) \propto (E - E_c)^{1/2}$ and using Boltzmann statistics $f(E) \approx \exp\left[-(E - E_F)/k_BT\right]$, show that the energy distribution of the electrons in the CB can be written as

$$n_x(x) = Cx^{1/2}\exp(-x)$$

where $x = (E - E_c)/k_BT$ is the electron energy in terms of k_BT measured from E_c and C is a constant at a given temperature (independent of E).

(b) Setting arbitrarily $C = 1$, plot $n_x(x)$ vs. x. Where is the maximum and what is the FWHM (full width at half maximum, that is, between half maximum points)? Is the use of $1.8k_BT$ for the half-maximum width correct?

(c) Show that the average electron energy in the CB is $(3/2)k_BT$, by using the definition of average,

$$x_{\text{average}} = \int_0^\infty x n_x dx \Big/ \int_0^\infty n_x dx$$

where the integration is from $x = 0$ (E_c) to say $x = 10$ (far away from E_c where $n_x \to 0$). You need to use a numerical integration.

(d) Show that the maximum in the energy distribution is at $x = (1/2)$ or at $E_{\max} = (1/2)k_BT$.

3.5 Intrinsic and doped GaAs The properties of GaAs are shown in Table 3.1. Calculate the intrinsic concentration and the intrinsic resistivity at room temperature (take as 300 K). Where is the Fermi level? Assuming the N_c and N_v scale as $T^{3/2}$, what would be the intrinsic concentration at 100°C? If this GaAs crystal is doped with 10^{17} donors cm^{-3} (such as Te), where is the new Fermi level and what is the resistivity of the sample? The drift motilities in GaAs are shown in Table 3.3.

TABLE 3.3 Ionized dopant impurities scatter carriers and reduce the drift mobility. The dependence of μ_e for electrons and μ_h for holes on the total ionized dopant concentration

Dopant concentration (cm^{-3})	0	10^{14}	10^{15}	10^{16}	10^{17}	10^{18}
GaAs, μ_e ($cm^2\ V^{-1}\ s^{-1}$)	8500	–	8000	7000	5000	2400
GaAs, μ_h ($cm^2\ V^{-1}\ s^{-1}$)	400	–	380	310	250	160
Si, μ_e ($cm^2\ V^{-1}\ s^{-1}$)	1450	1420	1370	1200	730	280
Si, μ_h ($cm^2\ V^{-1}\ s^{-1}$)	490	485	478	444	328	157

3.6 Electrons in GaAs Given that the electron effective mass m_e^* for the GaAs is $0.067m_e$, calculate the thermal velocity of the electrons in the CB of a nondegenerately doped GaAs at room temperature (300 K). If μ_e is the drift mobility of the electrons and τ_e the mean free time between electron scattering events (between electrons and lattice vibrations) and if $\mu_e = e\tau_e/m_e^*$, calculate τ_e, given $\mu_e = 8500\,cm^2\ V^{-1}\ s^{-1}$. Calculate the **drift velocity** $v_d = \mu_e E$ of the CB electrons in an applied field E of $10^5\ V\ m^{-1}$. What is your conclusion?

3.7 Extrinsic *n*-GaAs An *n*-type GaAs crystal is doped with 10^{16} donors cm^{-3} (such as Te), what are the electron and hole concentrations, and the conductivity? (See Table 3.3.)

3.8 Extrinsic *n*-Si A Si crystal has been doped *n*-type with $1 \times 10^{17}\,cm^{-3}$ phosphorus (P) donors. The electron drift mobility μ_e depends on the total concentration of ionized dopants N_{dopant}, as in Table 3.3, inasmuch as these ionized dopants scatter the electrons and thereby decrease their drift mobility. What is the conductivity of the sample? Where is the Fermi level with respect to the intrinsic crystal?

3.9 Compensation doping in *n*-type Si An *n*-type Si sample has been doped with 10^{16} phosphorus (P) atoms cm^{-3}. (a) What are the electron and hole concentrations? (b) Calculate the room temperature conductivity of the sample. (c) Where is the Fermi level with respect to E_{Fi}? (d) If we now dope the crystal with 10^{17} boron acceptors, what will be the electron and hole concentrations? (e) Where is the Fermi level with respect to E_{Fi}?

3.10 Free carrier absorption in semiconductors The attenuation of light due to the optical field drifting the free carriers is called **free carrier absorption**. As the free electrons in a semiconductor crystal are accelerated by the optical field, they eventually become scattered by lattice vibrations or impurities, and pass the energy absorbed from the radiation to lattice vibrations. In such cases, ε_r'' and the AC conductivity σ at the same frequency are related by

$$\varepsilon_r'' = \sigma/\varepsilon_o\omega \tag{P3.1}$$

We consider a semiconductor in which the free carriers are electrons (an *n*-type semiconductor). The AC conductivity σ in general is given by

$$\sigma = \sigma_o/(1 + j\omega\tau_e) \tag{P3.2}$$

where σ_o is the DC conductivity, ω is the angular frequency of light, and τ_e is the scattering time of the conduction electrons. σ decreases with frequency in Eq. (P3.2). Consider an *n*-type semiconductor. The free carriers are the electrons in the CB. If the drift mobility of the electrons is μ_e, then $\mu_e = e\tau_e/m_e^*$, where m_e^* is the effective mass of the electrons in the CB of the semiconductor.[28] The DC conductivity $\sigma_o = en\mu_e$, where n is the concentration of CB electrons. Show that the absorption coefficient due to free carrier absorption (due to the conductivity) when $\omega > 1/\tau$ is given by

$$\alpha = \left(\frac{\sigma_o}{nc\varepsilon_o\tau^2}\right)\left(\frac{1}{\omega^2}\right) = \left(\frac{e^3 n}{4\pi^2 nc^3\varepsilon_o m_e^{*2}\mu_e}\right)\lambda^2 \tag{P3.3}$$

[28] As we know, conduction in an *n*-type semiconductor occurs by the drift of free electrons inside the semiconductor crystal. The dopants donate electrons to the crystal, which are free within the crystal.

where n is the refractive index. What would you expect if you plotted α vs. λ^2? Consider a Si crystal doped with 10^{15} cm^{-3} donors. Estimate the free carrier absorption (in m^{-1} and dB m^{-1}) at 1.55 and at 5 μm. What is your conclusion? [Although Eq. (P3.3) is a highly simplified approximation to describing free carrier absorption, it nonetheless provides a rough estimate of its magnitude.]

3.11 GaAs *pn* junction Consider a GaAs *pn* junction that has the following properties: $N_a = 10^{16}$ cm^{-3} (p-side); $N_d = 10^{18}$ cm^{-3} (n-side); $B = 2.0 \times 10^{-16}$ m^3 s^{-1}; cross-sectional area $A = 1.5$ mm $\times$ 1.5 mm. Assume a long diode. What is the diode current due to diffusion in the neutral regions and recombination in the SCL at 300 K when the forward voltage across the diode is 0.8 V and then 1.1 V? (Use the drift motilities in Table 3.3 for calculating the diffusion coefficients through the Einstein relation.)

3.12 InP *pn* junction Consider an InP *pn* junction that has the following properties: $N_a = 10^{15}$ cm^{-3} (p-side); $N_d = 10^{17}$ cm^{-3} (n-side); using $B \approx 4 \times 10^{-16}$ m^3 s^{-1}; cross-sectional area $A = 1$ mm $\times$ 1 mm. Assume a long diode. What is the diode current due to diffusion in the neutral regions and recombination in the SCL at 300 K when the forward voltage across the diode is 0.70 and 0.9 V? The electron mobility in the p-side is about ~6000 cm^2 s^{-1} and the hole mobility on the n-side is roughly ~100 cm^2 s^{-1}. (See also Table 3.1 for n_i and ε_r.) Comment on the ideality factor of this InP *pn* junction.

3.13 Si *pn* junction Consider a long *pn* junction diode with an acceptor doping N_a of 10^{18} cm^{-3} on the p-side and donor concentration of N_d on the n-side. The diode is forward biased and has a voltage of 0.6 V across it. The diode cross-sectional area is 1 mm^2. The minority carrier recombination time, τ, depends on the total dopant concentration, N_{dopant}(cm^{-3}), through the following approximate empirical relation

$$\tau \approx (5 \times 10^{-7})/(1 + 2 \times 10^{-17} N_{\text{dopant}})$$

where τ is in seconds.

(a) Suppose that $N_d = 10^{15}$ cm^{-3}. Then the depletion layer extends essentially into the n-side and we have to consider minority carrier recombination time, τ_h, in this region. Calculate the diffusion and recombination contributions to the total diode current given $N_a = 10^{18}$ cm^{-3} and $N_d = 10^{15}$ cm^{-3}. Use Table 3.3 for μ_e and μ_h. What is your conclusion?

(b) Suppose that $N_d = N_a$. Then W extends equally to both sides and, further, $\tau_e = \tau_h$. Calculate the diffusion and recombination contributions to the diode current given $N_a = 10^{18}$ cm^{-3} and $N_d = 10^{18}$ cm^{-3}. Use Table 3.3 for μ_e and μ_h. What is your conclusion?

3.14 Injected minority carrier charge Consider a *pn* junction with heavier doping on the p-side. The injected minority carriers (holes) represent an *injected excess minority carrier charge* Q_h in the neutral region as shown in Figure 3.24 (a). (There is also excess majority carrier charge so the region is neutral.) Show that

$$Q = I\tau_h \text{ for a long diode and } Q = I\tau_t \text{ for a short diode}$$

in which τ_h is the hole lifetime and τ_t is the diffusion time, or the transit time of holes across the width of the neutral n-region, that is, $\tau_t = l_n^2/2D_h$. What is your conclusion?

3.15 High injection condition The Shockley equation for a *pn* junction under forward bias, as shown in Figure 3.16 (a), was derived by assuming low (weak) injection conditions, that is $p_n(0) \approx \Delta p_n(0) \ll n_{no}$ or N_d on the n-side. Show that when the injection is no longer weak, that is when $p_n(0) \approx n_{no} = N_d$, the applied voltage V reaches V_{SI} (strong injection) given by

$$V_{\text{SI}} = V_o - V_{\text{th}} \ln(N_a/N_d)$$

where V_{th} is the thermal voltage (k_BT/e). Calculate V_o and V_{SI} for a Si *pn* junction that has $N_a = 10^{18}$ cm^{-3} and $N_d = 10^{16}$ cm^{-3}. Can you use the Shockley equation when $V > V_{\text{SI}}$? What happens when $N_a = N_d$? What is your conclusion?

3.16 Heterostructure Consider a Type I heterostructure as shown in Figure 3.27.

(a) If $E_{g1} < E_{g2}$ and if χ_1 and χ_2 are the electron affinities of each semicondcutor, show that

$$\Delta E_c = \chi_1 - \chi_2 \quad \text{and} \quad \Delta E_v = E_{g2} - E_{g1} - \Delta E_c$$

(b) Using the data in Table 3.1, draw the energy band diagram of an *nP* junction between an n-type Ge and P-GaAs. Under forward bias, is it easier to inject electrons or holes?

(c) Draw the energy band diagram for a *pN* junction between p-type Ge and N-GaAs. Under forward bias, is it easier to inject electrons or holes?

3.17 Heterojunction *I–V* characteristics We use some of the data reported by Womac and Rediker (*J. Appl. Phys.*, *43*, 4130, 1972) for AlGaAs/GaAs *pn* heterojunction at 298 K (25°C). Sample A is an N^+p and sample B is a P^+n junction. The *I–V* data of interest are listed in Table 3.4. By a suitable plot find the ideality factor for each. What is your conclusion?

TABLE 3.4 *I–V* **data on two heterojunctions. First set is N^+p and the second set is a P^+n junction**

A: Heterojunction N^+p														
V	0.206 V	0.244	0.290	0.322	0.362	0.412	0.453	0.485	0.537	0.576	0.612	0.662	0.708	
I	1.03 nA	2.07	5.20	10.3	20.7	52.8	105	192	515	1.02 μA	2.03	4.89	10.1	
B: Heterojunction P^+n														
V	0.310 V	0.364	0.402	0.433	0.485	0.521	0.561	0.608	0.682	0.726	0.764	0.807	0.859	0.885
I	2.01 nA	4.91	9.79	19.0	49.5	96.1	194	466	1.96 μ A	5.02	9.75	19.5	50.6	99.5

3.18 AlGaAs LED emitter An AlGaAs LED emitter for in a local optical fiber network has the output spectrum shown in Figure 3.32 (b). It is designed for peak emission at about 822 nm at 25°C. (a) Why does the peak emission wavelength increase with temperature? (b) What is the bandgap of AlGaAs in this LED? (c) The bandgap, E_g, of the ternary $Al_xGa_{1-x}As$ alloys follows the empirical expression, $E_g(eV) = 1.424 + 1.266x + 0.266x^2$. What is the composition of the $Al_xGa_{1-x}As$ in this LED?

3.19 III–V compound semiconductors in optoelectronics Figure 3.52 represents the bandgap E_g and the lattice parameter a in a quaternary III–V alloy system. A line joining two points represents the changes in E_g and a with composition in a ternary alloy composed of the compounds at the ends of that line. For example, starting at GaAs point, $E_g = 1.42$ eV and $a = 0.565$ nm, E_g decreases and a increases as GaAs is alloyed with InAs, as we move

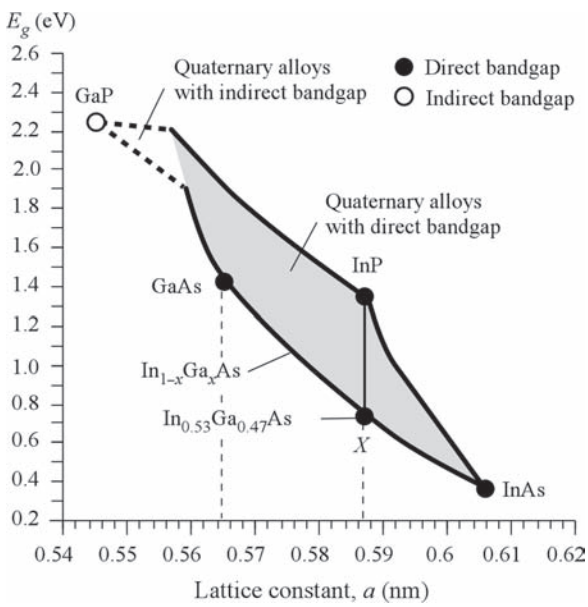
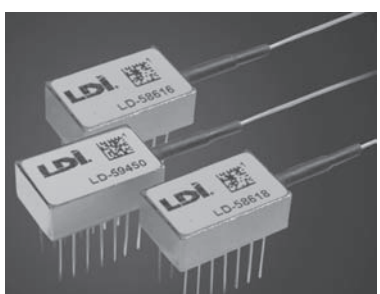


FIGURE 3.52 Bandgap energy E_g and lattice constant a for various III–V alloys of GaP, GaAs, InP, and InAs. A line represents a ternary alloy formed with compounds from the end points of the line. Solid lines are for direct bandgap alloys whereas dashed lines for indirect bandgap alloys. Regions between lines represent quaternary alloys. The line from X to InP represents quaternary alloys $In_{1-x}Ga_xAs_{1-y}P_y$ made from $In_{0.53}Ga_{0.47}As$ and InP, which are lattice-matched to InP.



InGaAsP 1300 nm LED emitters, each pigtailed to an optical fiber for use in optical communication modems and lower speed data/analog transmission systems. *(Courtesy of OSI Laser Diode, Inc.)*

along the line joining GaAs to InAs. Eventually at InAs, $E_g = 0.35\,\text{eV}$ and $a = 0.606\,\text{nm}$. Point X in Figure 3.52 is composed of InAs and GaAs and it is the ternary alloy $In_{1-x}Ga_xAs$. At X, $In_{0.53}Ga_{0.47}As$ (often called "in-gas" in telecom) has $E_g = 0.73\,\text{eV}$ and $a = 0.587\,\text{nm}$, which is the same a as that for InP. $In_{1-x}Ga_xAs$ at X is therefore lattice-matched to InP and can hence be grown on an InP substrate without creating defects at the interface.

Further, $In_{1-x}Ga_x$ As at X can be alloyed with InP to obtain a quaternary alloy[29] $In_{1-x}Ga_xAs_{1-y}P_y$ whose properties lie on the line joining X and InP and therefore all have the same lattice parameter as InP but different bandgap. Layers of $In_{1-x}Ga_xAs_{1-y}P_y$ with composition between X and InP can be grown epitaxially on an InP substrate by various techniques such as liquid phase epitaxy (LPE) or molecular beam epitaxy (MBE). The grey shaded area between the solid lines represents the possible values of E_g and a for the quaternary III–V alloy system in which the bandgap is direct and hence suitable for direct recombination. The compositions of the quaternary alloy lattice matched to InP follow the line from X to InP.

(a) Given that the $In_{1-x}Ga_x$ As at X is $In_{0.53}Ga_{0.47}As$ show that quaternary alloys $In_{1-x}Ga_xAs_{1-y}P_y$ are lattice matched to InP when $y = 1 - 2.13x$.

(b) The bandgap energy E_g, in eV for $In_{1-x}Ga_xAs_{1-y}P_y$ lattice-matched to InP is given by the empirical relation, $E_g(\text{eV}) = 0.75 + 0.46y + 0.14y^2$. Find the composition of the quaternary alloy suitable for an LED emitter operating at 1.30 μm.

3.20 Varshni equation and the change in the bandgap with temperature The Varshni equation describes the change in the energy bandgap E_g of a semiconductor with temperature T as given by Eq. (3.11.2) that is

$$E_g = E_{go} - AT^2/(B + T)$$

where E_{go} is E_g at 0 K, and A and B are constants. Show that

$$\frac{dE_g}{dT} = -\frac{AT(T + 2B)}{(B + T)^2} = -\frac{(E_{go} - E_g)}{T}\left(\frac{T + 2B}{T + B}\right)$$

For GaAs, $E_{go} = 1.519\,\text{eV}$, $A = 5.41 \times 10^{-4}\,\text{eV K}^{-1}$, $B = 204$ K. What is dE_g/dT for GaAs? Find the shift in the emitted wavelength from a GaAs LED per 1°C change at room temperature (300 K). Find the emission wavelength at 27°C and −30°C.

3.21 Dependence on the emission peak and linewidth on temperature Using the Varshni equation find the peak emission wavelength and the linewidth of the emission spectrum from an $In_{0.47}Ga_{0.53}As$ LED when it is cooled from 25°C to −25°C. You can use Eqs. (3.3.1) and (3.3.2). The Varshni constants for $In_{0.47}Ga_{0.53}As$ are $E_{go} = 0.814\,\text{eV}$, $A = 4.906 \times 10^{-4}\,\text{eV K}^{-1}$, $B = 301$ K.

3.22 LED Output Spectrum Given that the width of the relative light intensity vs. photon energy spectrum of an LED is typically around ~3 k_BT, calculate the spectral width in wavelength (nm) of LED emitters operating at 850 nm, 1310 nm, and 1550 nm?

3.23 Linewidth of LEDs Experiments carried out on various direct bandgap semiconductor LEDs give the output spectral linewidth (between half intensity points as listed in Table 3.5). From Figure 3.31 we know that a spread in the wavelength is related to a spread in the photon energy, $\Delta\lambda \approx (hc/E_{\text{ph}}^2)\Delta E_{\text{ph}}$ where $E_{\text{ph}} = h\upsilon$ is the photon energy. Suppose that we write $E_{\text{ph}} = hc/\lambda$ and $\Delta E_{\text{ph}} = \Delta(h\upsilon) \approx mk_BT$ where m is a numerical constant. Therefore,

$$\Delta\lambda \approx (mk_BT/hc)\lambda^2 \qquad \textbf{(P3.4)}$$

By appropriately plotting the data in Table 3.5, and assuming $T = 300\,\text{K}$, find m.

TABLE 3.5 Linewidth $\Delta\lambda_{1/2}$ between half points in the output spectrum (spectral intensity vs. wavelength) of eight LEDs using direct bandgap semiconductors

Material (Direct E_g)	AlGaAs	AlGaAs	AlGaAs	GaAs	GaAs	InGaAsP	InGaAsP	InGaAsP
Peak wavelength of emission (λ) nm	650	810	820	890	950	1150	1270	1500
$\Delta\lambda_{1/2}$ nm	22	36	40	50	55	90	110	150

[29] Some books have other formats for the chemical composition for example, $Ga_xIn_{1-x}As_yP_{1-y}$. The present notation $In_{1-x}Ga_xAs_{1-y}P_y$ was chosen to reflect the common vernacular for InGaAs (pronounced "in-gas").

Table 3.6 gives the linewidth $\Delta\lambda_{1/2}$ for various visible LEDs based on GaAsP. Radiative recombination is obtained by appropriately doping the material. Using $m = 3.0$ in Eq. (P3.4), $T = 300$ K, calculate the expected spectral width for each and compare with the experimental value. What is your conclusion?

TABLE 3.6 Linewidth $\Delta\lambda_{1/2}$ between half points in the output spectrum (intensity vs. wavelength) of various visible LEDs using GaAsP

Peak wavelength of emission (λ) nm	565	583	600	635
$\Delta\lambda_{1/2}$ nm	28	36	40	40
Color	Green	Yellow	Orange	Red
Material	GaP(N)	GaAsP(N)	GaAs (N)	GaAsP

3.24 Quantum wells We will consider a quantum well in a semiconductor heterostructure, and assume, for simplicity, an infinite potential energy quantum well with dimensions d along x, and D_y and D_z along y and z directions. The energy of an electron with respect to the bottom of the well is then given by

$$\Delta E = \frac{h^2 n^2}{8m_e^* d^2} + \frac{h^2 n_y^2}{8m_e^* D_y^2} + \frac{h^2 n_z^2}{8m_e^* D_z^2} \qquad \textbf{(P3.5)}$$

Energy in a quantum well

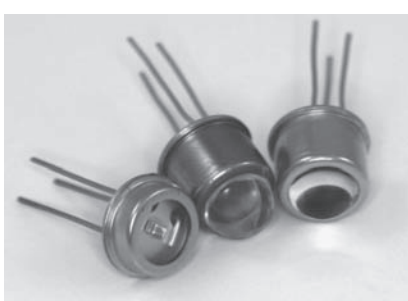
where n, n_y, and n_z are quantum numbers having the values 1, 2, 3, … Consider a well that has $d = 10$ nm, $D_x = D_y = 2\,\mu$m, and $m_e^* = 0.067m_e$. (a) Calculate the minimum energy. Which is the most significant contributing term in Eq. (P3.5) to the lowest energy? (b) What would n_y need to be to get the same energy as the first term? (c) What is the separation δE between the energy levels for motion in the y and z plane? What is your conclusion?

3.25 Energy levels in a quantum well Consider a GaAs QW sandwiched between two $Al_{0.40}Ga_{0.60}As$ layers. The barrier height ΔE_c is approximately 0.30 eV, the electron effective mass in the well is $0.067m_e$, and the width of the QW (d) is 8 nm. Calculate the energy levels ΔE_1 and ΔE_2 from the bottom of E_c, assuming an infinite PE well. Compare these with the calculations for a finite PE well that give $\Delta E_1 = 0.050$ eV and $\Delta E_2 = 0.197$ eV. What is your conclusion?

3.26 QW LED Linewidth Consider the QW-based InGaN (green to blue) and AlGaN (UV) LEDs whose peak emission wavelengths (λ_o) and linewidths (between half intensity points, $\Delta\lambda$) are listed in Table 3.7. For each LED, apply the linewidth equation $\Delta\lambda \approx (mk_BT/hc)\lambda_o^2$ at 300 K, and find m. Do you think this formula can be used for QW LEDs?

TABLE 3.7 Avago ASMT 1W series of InGaN green, cyan, blue, and royal blue LEDs and SETi AlGaN UV LED

LED	Green	Cyan	Blue	Royal blue	UV
λ_o	519	497	454	450	270
$\Delta\lambda$	31.25	29	22.5	20	12



UV emitting LEDs *(Courtesy of Sensor Electronic)*

3.27 External conversion efficiency The total light output power, called the radiant flux, from a number of commercial LEDs is listed in Table 3.8. The table also lists the emitted peak wavelength and the measured linewidth for each LED. For each device calculate the external quantum efficiency (%), EQE, and m in $\Delta\lambda \approx (mk_BT/hc)\lambda^2$. What is your conclusion?

TABLE 3.8 Characteristics of selected commercial LEDs covering the IR, blue, and UV

	IR, Hamamatsu L10823 (InGaAs)	IR, Hamamatsu L10822 (InGaAs)	IR, Osram SFH406 (GaAs)	IR, Vishay TSAL6100 (GaAlAs)	IR, Hamamatsu L1909 (AlGaAs)	Super Blue, Cree (InGaN)	UV, UVTOP 315, SETi (InGaN)
V_F (V)	0.9	1.1	1.25	1.35	1.4	3.8	5.5
I_F (mA)	50	50	40	100	50	30	20
P_o (mW)	2	3.1	7	35	10	1.15	0.6
λ_o (nm)	1650	1300	950	940	870	466	315
$\Delta\lambda$ (nm)	120	90	55	50	80	65	11

3.28 Light extraction efficiency of LEDs Extraction efficiency or ratio depends on the LED structure and the means that have been used to manipulate light to escape from the chip.

(a) Figure 3.53 (a) shows a bare LED chip in which the light from the active region is incident on the semiconductor–air interface. The refractive indices of the semiconductor and ambient are n_s and n_a respectively. Assume a GaN crystal for the semiconductor with a refractive index of 2.5. For normal incidence, what is the transmittance T from the semiconductor to the air? What is the critical angle θ_c at this interface? The fraction of output light from the active region within the emission cone defined by the critical angle θ_c has been evaluated to be $(1/2)[1 - \cos\theta_c]$. Taking into account T as well, estimate the extraction efficiency (EE) if the ambient is air, $n_a = 1$.

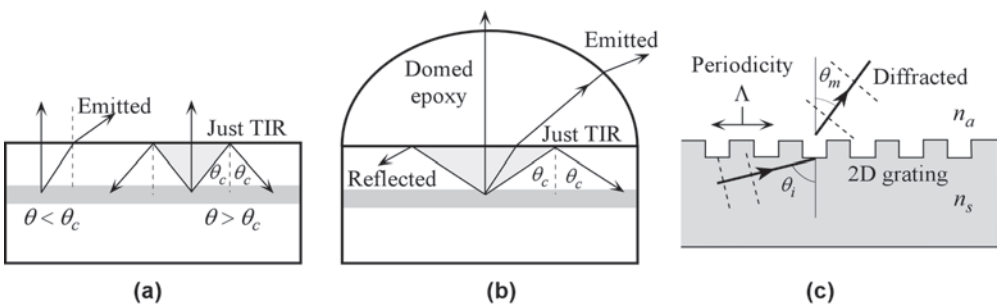


FIGURE 3.53 (a) Total internal reflection results in poor light extraction (b) Improvement in the light extraction by the use of a domed epoxy. (c) A two-dimensional diffraction grating (a photonic crystal) based surface enhances the extraction of light through Bragg diffraction.

(b) Figure 3.53 (b) shows an epoxy dome placed on top of the semiconductor. Suppose that the refractive index of the epoxy is 1.8. Calculate the new critical angle at the semiconductor–epoxy interface and estimate the new EE. What is the critical angle for light rays at the epoxy–air interface? How critical is it to have an exactly hemispherical surface?

(c) Figure 3.53 (c) shows an LED chip whose surface has been patterned to form a two-dimensional (2D) *photonic crystal*, which is a periodic modulation of the refractive index. With the right periodicity Λ, the photonic crystal will leak out the light trapped in the crystal. We will take a highly simplified view and treat the periodic variation as a grating. If θ_i is the angle of incidence, there is a diffracted beam at an angle θ_m, which satisfies the Bragg diffraction condition that involves two different media that is,

$$\Lambda(n_s \sin\theta_i - n_a \sin\theta_m) = m\lambda$$